Neutrino Oscillation on a Quantum Processor

A Thesis submitted for the completion of requirements for the degree of

Bachelor of Science (Research)

by

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Certificate

This is to certify that the Bachelor's project titled *"Neutrino Oscillation on a Quantum Processor"*, submitted by *Wangshitemjen Ozukum*
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Signature of the Supervisor: _____

Name of the Supervisor: Ranjan Laha

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I, *Wangshitemjen Ozukum*, declare that this project titled *"Neutrino Oscillation on a Quantum Processor"* and the work presented in it are entirely my own.

I confirm that:

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- Wherever I have consulted the published work of others, it has been properly acknowledged and cited.
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- I have acknowledged all major sources of assistance.
- Where the work is based on collaboration with others, I have clearly specified my contribution and that of others.

Name: Wangshitemjen ozukum

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Abstract

Neutrino oscillations are a quantum mechanical phenomenon where neutrinos change flavor as they propagate, implying non-zero masses and mixing. Simulating these oscillations on classical computers becomes computationally challenging when including effects such as matter interactions, decoherence, or sterile neutrinos, because the Hilbert space grows exponentially with the number of flavors. Quantum processors, which manipulate quantum states directly, offer a natural platform for such simulations. In this work, we map the two- and three-flavor oscillation dynamics to quantum circuits. The PMNS mixing matrix is implemented using single-qubit rotations and controlled-NOT gates, while time evolution in the mass basis is realized with phase gates. For three flavors, the evolution operator factorizes into a tensor product of single-qubit phase gates, avoiding the need for entanglement during propagation. Sterile neutrinos introduce a fourth mass eigenstate, which forces the evolution operator to become entangling; we provide a minimal circuit using two CNOTs and three phase gates to generate the required diagonal unitary. Decoherence is modelled via a Lindblad master equation and simulated by coupling the system to an ancilla qubit through controlled rotations, which damps off-diagonal density matrix elements. Matter effects are incorporated by redefining the mixing angle and squared-mass difference in the effective Hamiltonian. We present simulation results obtained on a quantum simulator, demonstrating that the quantum circuits faithfully reproduce the theoretical oscillation probabilities. This work shows that quantum processors can efficiently simulate complex neutrino phenomena and serve as a testbed for beyond-standard-model physics. [1, 2, 3, 4]

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Chapter 1

Introduction

Neutrinos are among the most abundant particles in the universe, yet they interact so weakly that they pass through matter almost undisturbed. One of the most intriguing properties of neutrinos is their ability to oscillate between different flavor states (ν_e, ν_μ, ν_τ) as they travel. This phenomenon, known as neutrino oscillation, implies that neutrinos have non-zero masses and that the flavor eigenstates are superpositions of mass eigenstates[1]. The theory of neutrino oscillations is well established and has been confirmed by numerous experiments [5]. However, simulating the full three-flavor dynamics including matter effects, decoherence, and potential sterile neutrinos becomes computationally intensive on classical computers because the Hilbert space dimension grows exponentially with the number of flavors[6]. For example, including sterile neutrinos increases the number of mass eigenstates to four, requiring a 4×4 mixing matrix and a correspondingly larger state space. Classical methods such as numerical integration of the Schrödinger equation or density-matrix evolution scale as $\mathcal{O}(N^2)$ for N eigenstates[6], which quickly becomes prohibitive when one considers time-dependent densities, non-standard interactions, or multiple active–sterile sectors.

Quantum processors, which operate on qubits and can directly realize unitary evolution and superposition, offer a natural framework for simulating such quantum systems. In recent years, there has been growing interest in using quantum computers to simulate neutrino oscillations [2, 7]. Unlike classical simulations that must store the full density matrix, quantum devices evolve the state naturally, potentially providing an exponential advantage in memory and sometimes in runtime for certain tasks. The goal of this thesis is to develop and implement quantum circuits that simulate two-flavor, three-flavor, and sterile neutrino oscillations, and to incorporate decoherence and matter effects. We show that these circuits can be executed on existing quantum hardware, reproducing the expected oscillation probabilities.

The thesis is organized as follows. Chapter 2 reviews the theory of neutrino oscilla-

tions, including the PMNS matrix, oscillation probability, and matter effects. Chapter 3 provides a brief introduction to quantum computing, focusing on the gates used later. Chapter 4 describes the simulation of two-flavor oscillations with a single qubit. Chapter 5 extends the simulation to three flavors using two qubits and presents the factorized time evolution. Chapter 6 introduces sterile neutrinos and the necessary entangling circuit. Chapter 7 covers the modelling of decoherence via an ancilla qubit. Chapter 8 discusses matter effects and their implementation. Chapter 10 presents the simulation results, and Chapter 11 concludes with a summary and future directions.

Chapter 2

Theory of Neutrino Oscillations

2.1 Flavor and Mass Eigenstates

Neutrinos are produced and detected in flavor eigenstates $|\nu_\alpha\rangle$ ($\alpha = e, \mu, \tau$), which are superpositions of mass eigenstates $|\nu_i\rangle$ ($i = 1, 2, 3$):

$$|\nu_\alpha\rangle = \sum_{i=1}^{3} U_{\alpha i}^{*} |\nu_i\rangle, \tag{2.1}$$

where U is the Pontecorvo–Maki–Nakagawa–Sakata (PMNS) mixing matrix[1]. Unitarity of U guarantees that the total probability is conserved. For three flavors, U is conventionally parameterized by three mixing angles $\theta_{12}, \theta_{13}, \theta_{23}$ and one CP-violating phase δ:

$$U = \begin{pmatrix} 1 & 0 & 0 \\ 0 & c_{23} & s_{23} \\ 0 & -s_{23} & c_{23} \end{pmatrix} \begin{pmatrix} c_{13} & 0 & s_{13} e^{-i\delta} \\ 0 & 1 & 0 \\ -s_{13} e^{i\delta} & 0 & c_{13} \end{pmatrix} \begin{pmatrix} c_{12} & s_{12} & 0 \\ -s_{12} & c_{12} & 0 \\ 0 & 0 & 1 \end{pmatrix}, \tag{2.2}$$

with $c_{ij} = \cos\theta_{ij}$, $s_{ij} = \sin\theta_{ij}$. In the standard parameterization, $\theta_{12} \approx 33.4^\circ$, $\theta_{23} \approx 42.3^\circ$, $\theta_{13} \approx 8.54^\circ$, and δ is currently constrained to be near 197° (normal ordering) [8]. The mass squared differences are $\Delta m^2_{21} = m_2^2 - m_1^2 \approx 7.5 \times 10^{-5}\,\mathrm{eV}^2$ and $|\Delta m^2_{31}| \approx 2.5 \times 10^{-3}\,\mathrm{eV}^2$, with the sign determining the mass ordering.

2.2 Oscillation Probability

A neutrino produced in a flavor eigenstate is a coherent superposition of mass eigenstates:

$$|\nu_\alpha\rangle = \sum_{i} U_{\alpha i}^{*} |\nu_i\rangle, \tag{2.3}$$

where U is the PMNS matrix. In vacuum, each mass eigenstate propagates with its own phase,

$$|\nu_i(t)\rangle = e^{-iE_i t}|\nu_i(0)\rangle. \tag{2.4}$$

For ultra-relativistic neutrinos,

$$E_i \approx E + \frac{m_i^2}{2E}, \tag{2.5}$$

so after traveling a distance L (with $t \approx L$ in natural units), the state becomes

$$|\nu_\alpha(L)\rangle = \sum_i U_{\alpha i}^* e^{-i\frac{m_i^2 L}{2E}}|\nu_i\rangle. \tag{2.6}$$

To find the amplitude for detecting flavor β, we project onto $\langle\nu_\beta|$:

$$\mathcal{A}_{\alpha\to\beta}(L) = \langle\nu_\beta|\nu_\alpha(L)\rangle = \sum_i U_{\beta i}U_{\alpha i}^* e^{-i\frac{m_i^2 L}{2E}}. \tag{2.7}$$

Therefore, the oscillation probability is[1]

$$P_{\alpha\to\beta}(L) = \left|\sum_i U_{\beta i}U_{\alpha i}^* e^{-i\frac{m_i^2 L}{2E}}\right|^2. \tag{2.8}$$

Only phase differences are physically relevant. Defining

$$\Delta m_{ij}^2 = m_i^2 - m_j^2, \qquad \Delta_{ij} \equiv \frac{\Delta m_{ij}^2 L}{2E}, \tag{2.9}$$

the oscillatory behavior is governed by the relative phases between different mass eigenstates.

2.2.1 Two-Flavor Oscillation Probability

For two flavors, the mixing matrix can be written as

$$U = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}. \tag{2.10}$$

Thus,

$$|\nu_e\rangle = \cos\theta|\nu_1\rangle + \sin\theta|\nu_2\rangle, \qquad |\nu_\mu\rangle = -\sin\theta|\nu_1\rangle + \cos\theta|\nu_2\rangle. \tag{2.11}$$

Suppose the neutrino is produced as $|\nu_e\rangle$. After propagation,

$$|\nu_e(L)\rangle = \cos\theta\, e^{-i\phi_1}|\nu_1\rangle + \sin\theta\, e^{-i\phi_2}|\nu_2\rangle, \tag{2.12}$$

where

$$\phi_i = \frac{m_i^2 L}{2E}. \tag{2.13}$$

The survival amplitude is

$$\mathcal{A}_{e\to e} = \langle \nu_e | \nu_e(L) \rangle = \cos^2\theta\, e^{-i\phi_1} + \sin^2\theta\, e^{-i\phi_2}. \tag{2.14}$$

Factoring out the overall phase $e^{-i\phi_1}$,

$$\mathcal{A}_{e\to e} = e^{-i\phi_1}\left(\cos^2\theta + \sin^2\theta\, e^{-i(\phi_2-\phi_1)}\right). \tag{2.15}$$

Since an overall phase does not affect the probability,

$$P_{e\to e} = \left|\cos^2\theta + \sin^2\theta\, e^{-i\Delta\phi}\right|^2, \qquad \Delta\phi = \phi_2 - \phi_1 = \frac{\Delta m^2 L}{2E}, \tag{2.16}$$

with $\Delta m^2 \equiv m_2^2 - m_1^2$. Expanding,

$$P_{e\to e} = \cos^4\theta + \sin^4\theta + 2\sin^2\theta\cos^2\theta\cos\Delta\phi \tag{2.17}$$

$$= 1 - 2\sin^2\theta\cos^2\theta\,(1-\cos\Delta\phi). \tag{2.18}$$

Using

$$1 - \cos\Delta\phi = 2\sin^2\left(\frac{\Delta\phi}{2}\right), \qquad \sin^2 2\theta = 4\sin^2\theta\cos^2\theta, \tag{2.19}$$

we obtain

$$P_{e\to e} = 1 - \sin^2 2\theta\, \sin^2\left(\frac{\Delta m^2 L}{4E}\right). \tag{2.20}$$

Now consider the appearance channel. The transition amplitude is

$$\mathcal{A}_{e\to\mu} = \langle \nu_\mu | \nu_e(L) \rangle = \sin\theta\cos\theta\left(e^{-i\phi_2} - e^{-i\phi_1}\right). \tag{2.21}$$

Therefore,

$$P_{e\to\mu} = \sin^2\theta\cos^2\theta\left|e^{-i\phi_2} - e^{-i\phi_1}\right|^2. \tag{2.22}$$

Using

$$\left|1 - e^{-i\Delta\phi}\right|^2 = 4\sin^2\left(\frac{\Delta\phi}{2}\right), \tag{2.23}$$

we get

$$P_{e\to\mu} = \sin^2 2\theta\, \sin^2\left(\frac{\Delta m^2 L}{4E}\right). \tag{2.24}$$

Hence, in the two-flavor case[1],

$$P_{e\to e} + P_{e\to\mu} = 1. \tag{2.25}$$

2.2.2 Three-Flavor Oscillation Probability

For three flavors, the amplitude is

$$\mathcal{A}_{\alpha\to\beta} = \sum_{i=1}^{3} U_{\beta i} U_{\alpha i}^{*} e^{-i\frac{m_i^2 L}{2E}}, \tag{2.26}$$

so the probability is

$$P_{\alpha\to\beta} = \left| \sum_{i=1}^{3} U_{\beta i} U_{\alpha i}^{*} e^{-i\frac{m_i^2 L}{2E}} \right|^2 . \tag{2.27}$$

Expanding the modulus squared,

$$P_{\alpha\to\beta} = \sum_{i,j} U_{\beta i} U_{\alpha i}^{*} U_{\beta j}^{*} U_{\alpha j} e^{-i\frac{(m_i^2 - m_j^2)L}{2E}}. \tag{2.28}$$

Separating diagonal and off-diagonal terms,

$$P_{\alpha\to\beta} = \sum_{i} |U_{\beta i}|^2 |U_{\alpha i}|^2 + \sum_{i\neq j} U_{\beta i} U_{\alpha i}^{*} U_{\beta j}^{*} U_{\alpha j} e^{-i\Delta_{ij}}. \tag{2.29}$$

Grouping terms with (i, j) and (j, i) together,

$$P_{\alpha\to\beta} = \sum_{i} |U_{\beta i}|^2 |U_{\alpha i}|^2 + 2\sum_{i<j} \Re\left[U_{\beta i} U_{\alpha i}^{*} U_{\beta j}^{*} U_{\alpha j} e^{-i\Delta_{ij}} \right]. \tag{2.30}$$

Define

$$X_{ij}^{\alpha\beta} = U_{\alpha i}^{*} U_{\beta i} U_{\alpha j} U_{\beta j}^{*}. \tag{2.31}$$

Then the probability can be written as

$$P_{\alpha\to\beta} = \sum_{i} |U_{\beta i}|^2 |U_{\alpha i}|^2 + 2\sum_{i<j} \Re(X_{ij}^{\alpha\beta}) \cos\Delta_{ij} + 2\sum_{i<j} \Im(X_{ij}^{\alpha\beta}) \sin\Delta_{ij}. \tag{2.32}$$

Using $\cos\Delta_{ij} = 1 - 2\sin^2(\Delta_{ij}/2)$ and unitarity of the PMNS matrix, this becomes the standard vacuum formula

$$P_{\alpha\to\beta} = \delta_{\alpha\beta} - 4\sum_{i<j} \Re\left[U_{\alpha i}^{*} U_{\beta i} U_{\alpha j} U_{\beta j}^{*} \right] \sin^2\left(\frac{\Delta m_{ij}^2 L}{4E} \right) + 2\sum_{i<j} \Im\left[U_{\alpha i}^{*} U_{\beta i} U_{\alpha j} U_{\beta j}^{*} \right] \sin\left(\frac{\Delta m_{ij}^2 L}{2E} \right). \tag{2.33}$$

The first oscillatory term is CP-even, while the second is CP-odd and appears only if the PMNS matrix has a non-zero complex phase. Thus the three-flavor case naturally allows CP violation in neutrino oscillations.

For the survival probability, $\beta = \alpha$, so the imaginary term vanishes and we obtain

$$P_{\alpha\to\alpha} = 1 - 4\sum_{i<j} |U_{\alpha i}|^2 |U_{\alpha j}|^2 \sin^2\left(\frac{\Delta m^2_{ij} L}{4E}\right). \tag{2.34}$$

For example,

$$\begin{aligned} P_{e\to e} = 1 &- 4|U_{e1}|^2|U_{e2}|^2 \sin^2\left(\frac{\Delta m^2_{21} L}{4E}\right) - 4|U_{e1}|^2|U_{e3}|^2 \sin^2\left(\frac{\Delta m^2_{31} L}{4E}\right) \\ &- 4|U_{e2}|^2|U_{e3}|^2 \sin^2\left(\frac{\Delta m^2_{32} L}{4E}\right). \end{aligned} \tag{2.35}$$

Similarly,

$$\begin{aligned} P_{\mu\to\mu} = 1 &- 4|U_{\mu1}|^2|U_{\mu2}|^2 \sin^2\left(\frac{\Delta m^2_{21} L}{4E}\right) - 4|U_{\mu1}|^2|U_{\mu3}|^2 \sin^2\left(\frac{\Delta m^2_{31} L}{4E}\right) \\ &- 4|U_{\mu2}|^2|U_{\mu3}|^2 \sin^2\left(\frac{\Delta m^2_{32} L}{4E}\right). \end{aligned} \tag{2.36}$$

Therefore, the two-flavor formula appears as a special limit of the more general three-flavor expression when only one mass-squared splitting and one effective mixing angle dominate. [1]

2.3 Matter Effects

When neutrinos propagate through matter, coherent forward scattering with electrons modifies the effective Hamiltonian. In the two-flavor case, the Hamiltonian in the flavor basis becomes

$$H = \begin{pmatrix} V - \frac{\Delta m^2}{4E}\cos 2\theta & \frac{\Delta m^2}{4E}\sin 2\theta \\ \frac{\Delta m^2}{4E}\sin 2\theta & \frac{\Delta m^2}{4E}\cos 2\theta \end{pmatrix}, \tag{2.37}$$

where $V = \sqrt{2} G_F n_e$ is the matter potential (G_F the Fermi constant, n_e the electron number density)[9, 10, 1]. Diagonalizing this Hamiltonian yields a matter-modified mixing angle θ_m and squared-mass difference Δm^2_m:

$$\tan 2\theta_m = \frac{\Delta m^2 \sin 2\theta}{\Delta m^2 \cos 2\theta - 2EV}, \tag{2.38}$$

$$\Delta m^2_m = \sqrt{(\Delta m^2 \cos 2\theta - 2EV)^2 + (\Delta m^2 \sin 2\theta)^2}. \tag{2.39}$$

The oscillation probability in matter then takes the same form as in vacuum, with θ and Δm^2 replaced by θ_m and Δm^2_m. The matter potential can lead to resonant enhancement

(the Mikheyev–Smirnov–Wolfenstein effect) when $2EV = \Delta m^2 \cos 2\theta$, which is of great importance for solar and supernova neutrinos[9, 10, 1].

Chapter 3

Quantum Computing Background

3.1 Qubits and Quantum Gates

A qubit is a two-level quantum system, whose state is a vector in $\mathbb{C}^2$: $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ with $|\alpha|^2 + |\beta|^2 = 1$. Quantum gates are unitary operators acting on qubits. The most common single-qubit gates are the Pauli matrices[6]:

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \tag{3.1}$$

and rotation gates $R_x(\theta) = e^{-i\theta X/2}$, $R_y(\theta) = e^{-i\theta Y/2}$, $R_z(\theta) = e^{-i\theta Z/2}$[6].

3.2 Decomposition of a 2×2 complex matrix.

Any 2×2 complex matrix $A \in \mathbb{C}^{2\times 2}$ can be written as a linear combination of the identity matrix and the Pauli matrices[6]:

$$A = c_0 I + c_1 \sigma_x + c_2 \sigma_y + c_3 \sigma_z, \tag{3.2}$$

where $c_0, c_1, c_2, c_3 \in \mathbb{C}$.

Explicitly, for

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \tag{3.3}$$

the coefficients are given by

$$A = \frac{a+d}{2} I + \frac{b+c}{2} \sigma_x + \frac{c-b}{2i} \sigma_y + \frac{a-d}{2} \sigma_z. \tag{3.4}$$

Equivalently, the coefficients can be obtained using the trace relation:

$$c_\mu = \frac{1}{2}\mathrm{Tr}(\sigma_\mu A), \quad \mu = 0, 1, 2, 3, \tag{3.5}$$

where $\sigma_0 \equiv I$.

3.3 Most general single-qubit unitary.

A general 2×2 complex matrix can be written as

$$U = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \qquad a, b, c, d \in \mathbb{C}. \tag{3.6}$$

Imposing unitarity,

$$U^\dagger U = I, \tag{3.7}$$

we obtain

$$|a|^2 + |c|^2 = 1, \tag{3.8}$$
$$|b|^2 + |d|^2 = 1, \tag{3.9}$$
$$a^* b + c^* d = 0. \tag{3.10}$$

Thus, the columns of U form an orthonormal set.

Parametrization of the first column. We parametrize the first column as

$$\begin{pmatrix} a \\ c \end{pmatrix} = \begin{pmatrix} e^{i\alpha} \cos\frac{\theta}{2} \\ e^{i\beta} \sin\frac{\theta}{2} \end{pmatrix}, \tag{3.11}$$

where $\theta \in [0, \pi]$ and $\alpha, \beta \in \mathbb{R}$.

Construction of the second column. A normalized vector orthogonal to the first column is

$$\begin{pmatrix} b \\ d \end{pmatrix} = \begin{pmatrix} -e^{i(\alpha+\lambda)} \sin\frac{\theta}{2} \\ e^{i(\beta+\lambda)} \cos\frac{\theta}{2} \end{pmatrix}, \tag{3.12}$$

where $\lambda \in \mathbb{R}$ is an arbitrary phase.

General unitary matrix. Thus, the most general unitary matrix can be written as

$$U = \begin{pmatrix} e^{i\alpha}\cos\frac{\theta}{2} & -e^{i(\alpha+\lambda)}\sin\frac{\theta}{2} \\ e^{i\beta}\sin\frac{\theta}{2} & e^{i(\beta+\lambda)}\cos\frac{\theta}{2} \end{pmatrix}. \tag{3.13}$$

Removing the global phase. Factoring out a global phase $e^{i\alpha}$ and defining

$$\phi = \beta - \alpha, \tag{3.14}$$

we obtain

$$U = e^{i\alpha}\begin{pmatrix} \cos\frac{\theta}{2} & -e^{i\lambda}\sin\frac{\theta}{2} \\ e^{i\phi}\sin\frac{\theta}{2} & e^{i(\phi+\lambda)}\cos\frac{\theta}{2} \end{pmatrix}. \tag{3.15}$$

Dropping the global phase (physically irrelevant), the most general single-qubit unitary is [6, 11]

$$U(\theta,\phi,\lambda) = \begin{pmatrix} \cos\frac{\theta}{2} & -e^{i\lambda}\sin\frac{\theta}{2} \\ e^{i\phi}\sin\frac{\theta}{2} & e^{i(\phi+\lambda)}\cos\frac{\theta}{2} \end{pmatrix}. \tag{3.16}$$

The phase gate $S(\phi)$ is a special case: $S(\phi) = U_3(0,0,\phi) = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\phi} \end{pmatrix}$.

3.4 Controlled-NOT Gates

We work in the standard computational basis:

$$\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}.$$

CNOT with control on qubit 1 and target on qubit 2 (CNOT_{12})

The action of the gate is:

$$|00\rangle \to |00\rangle \tag{3.17}$$
$$|01\rangle \to |01\rangle \tag{3.18}$$
$$|10\rangle \to |11\rangle \tag{3.19}$$
$$|11\rangle \to |10\rangle \tag{3.20}$$

The matrix representation is:

$$\mathrm{CNOT}_{12} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

It can also be written in tensor product form as:

$$\mathrm{CNOT}_{12} = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes X.$$

CNOT with control on qubit 2 and target on qubit 1 (CNOT_{21})

The action of the gate is:

$$|00\rangle \to |00\rangle \tag{3.21}$$
$$|01\rangle \to |11\rangle \tag{3.22}$$
$$|10\rangle \to |10\rangle \tag{3.23}$$
$$|11\rangle \to |01\rangle \tag{3.24}$$

The matrix representation is:

$$\mathrm{CNOT}_{21} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}.$$

In tensor product form:

$$\mathrm{CNOT}_{21} = I \otimes |0\rangle\langle 0| + X \otimes |1\rangle\langle 1|.$$

[6, 11]

3.5 Mapping Neutrino Dynamics to Circuits

The oscillation process can be broken into three steps[2]:

1. At production, the flavor state is converted to the mass basis by applying $U_{\text{PMNS}}^{\dagger}$.
2. The mass eigenstates evolve freely, acquiring phases $e^{-i\frac{m_i^2 L}{2E}}$.
3. At detection, the mass basis is converted back to flavor basis by applying U_{PMNS}.

Thus the overall evolution operator is

$$U_{\text{total}} = U_{\text{PMNS}}\, U_{\text{prop}}\, U_{\text{PMNS}}^{\dagger}, \tag{3.25}$$

where U_{prop} is diagonal in the mass basis. All operations are unitary and can be decomposed into elementary gates. For N flavors, the Hilbert space dimension is N, and we embed it into $\lceil \log_2 N \rceil$ qubits[6]. Because the number of qubits grows only logarithmically with N, this mapping is exponentially more efficient than storing the full density matrix classically.

Chapter 4

Two-Flavor Oscillation on a Quantum Processor

4.1 Two-Flavor Time Evolution as a Phase Gate

For a two-flavor system, the mass basis may be identified with the single-qubit computational basis,

$$|0\rangle \equiv |\nu_1\rangle, \qquad |1\rangle \equiv |\nu_2\rangle.$$

In the ultra-relativistic limit,

$$E_i \simeq E + \frac{m_i^2}{2E}.$$

After dropping the common energy term, the effective Hamiltonian in the mass basis becomes

$$H = \frac{1}{2E}\begin{pmatrix} m_1^2 & 0 \\ 0 & m_2^2 \end{pmatrix}.$$

This Hamiltonian can be written in Pauli form as [6]

$$H = \alpha I + \beta Z,$$

where

$$\alpha = \frac{m_1^2 + m_2^2}{4E}, \qquad \beta = \frac{m_1^2 - m_2^2}{4E}.$$

The time-evolution operator is

$$U(t) = e^{-iHt} = e^{-i\alpha t I} e^{-i\beta t Z}.$$

The factor $e^{-i\alpha t I}$ is a global phase and does not affect oscillation probabilities, so the

physical evolution is

$$U(t) = e^{-i\beta t Z}.$$

Using[6]

$$e^{-i\theta Z} = \begin{pmatrix} e^{-i\theta} & 0 \\ 0 & e^{i\theta} \end{pmatrix},$$

and removing an overall phase, one obtains

$$U(t) = \begin{pmatrix} 1 & 0 \\ 0 & e^{-i\frac{\Delta m^2 t}{2E}} \end{pmatrix},$$

where $\Delta m^2 = m_2^2 - m_1^2$. Defining

$$\phi = \frac{\Delta m^2 t}{2E\hbar},$$

the evolution operator becomes

$$U(t) = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\phi} \end{pmatrix} \equiv S(\phi).$$

Thus, in the two-flavor case, neutrino propagation in the mass basis is implemented by a single-qubit phase gate.

[2, 8, 12]

4.2 Removal of the ϕ and λ phases by field rephasing

In the two-flavor case, the most general single-qubit unitary gate used in the quantum-circuit construction is the three-parameter U_3 gate, given by [2]

$$U_3(\Theta, \phi, \lambda) = \begin{pmatrix} \cos\frac{\Theta}{2} & -e^{i\lambda}\sin\frac{\Theta}{2} \\ e^{i\phi}\sin\frac{\Theta}{2} & e^{i(\phi+\lambda)}\cos\frac{\Theta}{2} \end{pmatrix}. \tag{4.1}$$

This matrix may be factorized into three parts:

$$U_3(\Theta, \phi, \lambda) = \underbrace{\begin{pmatrix} 1 & 0 \\ 0 & e^{i\phi} \end{pmatrix}}_{P_\phi} \underbrace{\begin{pmatrix} \cos\frac{\Theta}{2} & -\sin\frac{\Theta}{2} \\ \sin\frac{\Theta}{2} & \cos\frac{\Theta}{2} \end{pmatrix}}_{R(\Theta)} \underbrace{\begin{pmatrix} 1 & 0 \\ 0 & e^{i\lambda} \end{pmatrix}}_{P_\lambda}. \tag{4.2}$$

Indeed, multiplying these matrices gives

$$P_\phi\, R(\Theta)\, P_\lambda = \begin{pmatrix} \cos\frac{\Theta}{2} & -e^{i\lambda}\sin\frac{\Theta}{2} \\ e^{i\phi}\sin\frac{\Theta}{2} & e^{i(\phi+\lambda)}\cos\frac{\Theta}{2} \end{pmatrix} = U_3(\Theta,\phi,\lambda). \tag{4.3}$$

Thus, the phase ϕ appears as a left diagonal phase matrix, the angle Θ appears in the central real rotation matrix, and the phase λ appears as a right diagonal phase matrix.

Now write the relation between flavor and mass eigenstates as

$$\begin{pmatrix} \nu_e \\ \nu_\mu \end{pmatrix} = U_3(\Theta,\phi,\lambda) \begin{pmatrix} \nu_1 \\ \nu_2 \end{pmatrix}. \tag{4.4}$$

Using the above factorization,

$$\begin{pmatrix} \nu_e \\ \nu_\mu \end{pmatrix} = P_\phi\, R(\Theta)\, P_\lambda \begin{pmatrix} \nu_1 \\ \nu_2 \end{pmatrix}. \tag{4.5}$$

Multiplying from the left by P_ϕ^{-1}, one obtains

$$\begin{pmatrix} \nu_e \\ e^{-i\phi}\nu_\mu \end{pmatrix} = R(\Theta) \begin{pmatrix} \nu_1 \\ e^{i\lambda}\nu_2 \end{pmatrix}. \tag{4.6}$$

Therefore, if one redefines the states as

$$\nu'_\mu = e^{-i\phi}\nu_\mu, \qquad \nu'_2 = e^{i\lambda}\nu_2, \tag{4.7}$$

then the mixing relation becomes simply

$$\begin{pmatrix} \nu_e \\ \nu'_\mu \end{pmatrix} = R(\Theta) \begin{pmatrix} \nu_1 \\ \nu'_2 \end{pmatrix}. \tag{4.8}$$

Hence, the phases ϕ and λ may be absorbed into field redefinitions. Therefore, for two-flavor neutrino oscillations one may, without loss of generality, set [5]

$$\phi = 0, \qquad \lambda = 0.$$

The general U_3 gate then reduces to a real rotation,

$$U_3(\Theta,0,0) = \begin{pmatrix} \cos\frac{\Theta}{2} & -\sin\frac{\Theta}{2} \\ \sin\frac{\Theta}{2} & \cos\frac{\Theta}{2} \end{pmatrix}.$$

Upon identifying $\Theta = 2\theta$, this becomes the standard two-flavor PMNS matrix,

$$U_{\mathrm{PMNS}}^{2\times 2} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}.$$

Thus, in the two-flavor oscillation framework, only the mixing angle θ is relevant for oscillation probabilities.

4.3 The Hermitian conjugate of the PMNS matrix

The two-flavor PMNS† matrix is

$$U_{\mathrm{PMNS}}^{\dagger} = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}. \tag{4.9}$$

This is a rotation in the x–z plane, and can be realized by $U_3(-2\theta, 0, 0)$. In the quantum circuit, we apply $U_{\mathrm{PMNS}}^{\dagger} = U_3(-2\theta, 0, 0)$ at the production step[2].

4.4 Two-flavor electron-neutrino disappearance and its quantum-circuit analog

In the two-flavor case we identify the computational basis with the mass basis[2],

$$|0\rangle \equiv |\nu_1\rangle, \qquad |1\rangle \equiv |\nu_2\rangle. \tag{4.10}$$

The two-flavor PMNS matrix is

$$U_{\mathrm{PMNS}} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}, \qquad U_{\mathrm{PMNS}}^{\dagger} = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}. \tag{4.11}$$

The electron-flavor state is obtained from the mass basis as

$$|\nu_e(0)\rangle = U_{\mathrm{PMNS}}^{\dagger}|0\rangle = \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix} = \cos\theta\,|\nu_1\rangle + \sin\theta\,|\nu_2\rangle. \tag{4.12}$$

In the mass basis, time evolution is diagonal:

$$U(t) = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\phi} \end{pmatrix}, \qquad \phi = \frac{\Delta m^2\, t}{2E\hbar}. \tag{4.13}$$

Therefore,

$$|\nu_e(t)\rangle = U(t)|\nu_e(0)\rangle = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\phi} \end{pmatrix} \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix} = \begin{pmatrix} \cos\theta \\ e^{i\phi}\sin\theta \end{pmatrix}. \tag{4.14}$$

The electron-neutrino survival amplitude is

$$A_{ee}(t) = \langle \nu_e(0)|\nu_e(t)\rangle. \tag{4.15}$$

Using

$$\langle \nu_e(0)| = \begin{pmatrix} \cos\theta & \sin\theta \end{pmatrix}, \tag{4.16}$$

we obtain

$$A_{ee}(t) = \begin{pmatrix} \cos\theta & \sin\theta \end{pmatrix} \begin{pmatrix} \cos\theta \\ e^{i\phi}\sin\theta \end{pmatrix} \tag{4.17}$$

$$= \cos^2\theta + e^{i\phi}\sin^2\theta. \tag{4.18}$$

Hence the survival probability is

$$P_{ee}(t) = |A_{ee}(t)|^2 = \left|\cos^2\theta + e^{i\phi}\sin^2\theta\right|^2. \tag{4.19}$$

Expanding this expression,

$$P_{ee}(t) = \left(\cos^2\theta + e^{i\phi}\sin^2\theta\right)\left(\cos^2\theta + e^{-i\phi}\sin^2\theta\right) \tag{4.20}$$

$$= \cos^4\theta + \sin^4\theta + 2\cos^2\theta\sin^2\theta\cos\phi \tag{4.21}$$

$$= 1 - \sin^2(2\theta)\sin^2\left(\frac{\phi}{2}\right). \tag{4.22}$$

On a quantum processor, we do not literally implement the bra $\langle \nu_e(0)|$. Instead, we prepare the state $|0\rangle$, construct the flavor state with $U^\dagger_{\text{PMNS}}$, apply the propagation gate $U(t)$, rotate back with U_{PMNS}, and then measure in the computational basis. Thus the relevant circuit is

$$|0\rangle \xrightarrow{U^\dagger_{\text{PMNS}}} |\nu_e\rangle \xrightarrow{U(t)} |\nu_e(t)\rangle \xrightarrow{U_{\text{PMNS}}} \text{measure}. \tag{4.23}$$

The full matrix action on the initial computational basis state is

$$U_{\text{PMNS}}\,U(t)\,U^\dagger_{\text{PMNS}}|0\rangle = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & e^{i\phi} \end{pmatrix} \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}. \tag{4.24}$$

Carrying out the multiplication gives

$$U_{\text{PMNS}}\, U(t)\, U_{\text{PMNS}}^{\dagger}|0\rangle = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} \cos\theta \\ e^{i\phi}\sin\theta \end{pmatrix} \tag{4.25}$$

$$= \begin{pmatrix} \cos^2\theta + e^{i\phi}\sin^2\theta \\ \sin\theta\cos\theta\,(e^{i\phi}-1) \end{pmatrix}. \tag{4.26}$$

Therefore the amplitude for measuring the final state $|0\rangle$ is

$$\langle 0|U_{\text{PMNS}}\, U(t)\, U_{\text{PMNS}}^{\dagger}|0\rangle = \cos^2\theta + e^{i\phi}\sin^2\theta, \tag{4.27}$$

and the corresponding measurement probability is

$$P(0) = \left|\langle 0|U_{\text{PMNS}}\, U(t)\, U_{\text{PMNS}}^{\dagger}|0\rangle\right|^2 = \left|\cos^2\theta + e^{i\phi}\sin^2\theta\right|^2. \tag{4.28}$$

Hence the quantum-circuit measurement probability is exactly equal to the electron-neutrino survival probability,

$$P(0) = P_{ee}(t). \tag{4.29}$$

This shows explicitly how two-flavor neutrino oscillation is mapped to a single-qubit quantum circuit.

4.5 Full Circuit

The complete circuit for two-flavor oscillations is therefore[2]:

$$\underbrace{U_3(2\theta,0,0)^{\dagger}}_{U_{\text{PMNS}}^{\dagger}} \quad S(\phi) \quad \underbrace{U_3(2\theta,0,0)}_{U_{\text{PMNS}}}.$$



Figure 4.1: This Circuit gives the $\nu_e \to \nu_e$ probability and $\nu_e \to \nu_\mu$ probability

We implemented this circuit on both simulators and real quantum processors. The

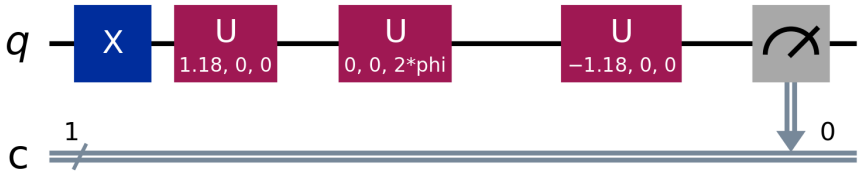


Figure 4.2: This circuit gives the $\nu_\mu \to \nu_\mu$ and $\nu_\mu \to \nu_e$ probability

results of the ν_e survival probabilities and the $\nu_e \to \nu_\mu$ appearance probabilities are shown in Figures 4.3 and 4.4, respectively. The experimental points were obtained by measuring the final state after many circuit repetitions and applying the appropriate measurement mapping.

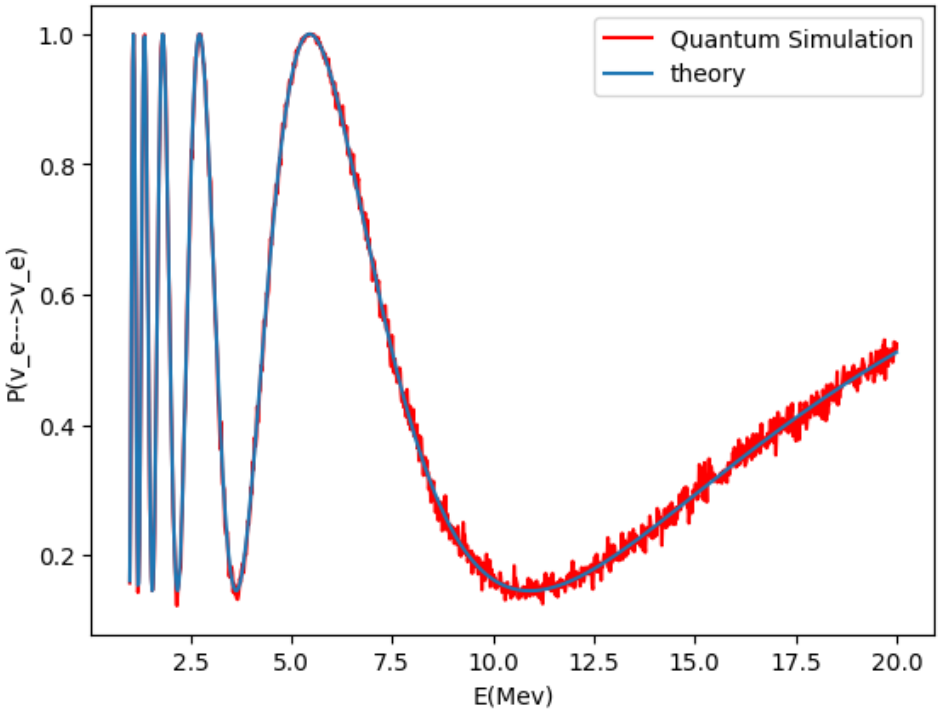


Figure 4.3: Two-flavor $\nu_e \to \nu_e$ survival probability.The solid blue curve is the theoretical prediction, and the red curve is the quantum simulation results.

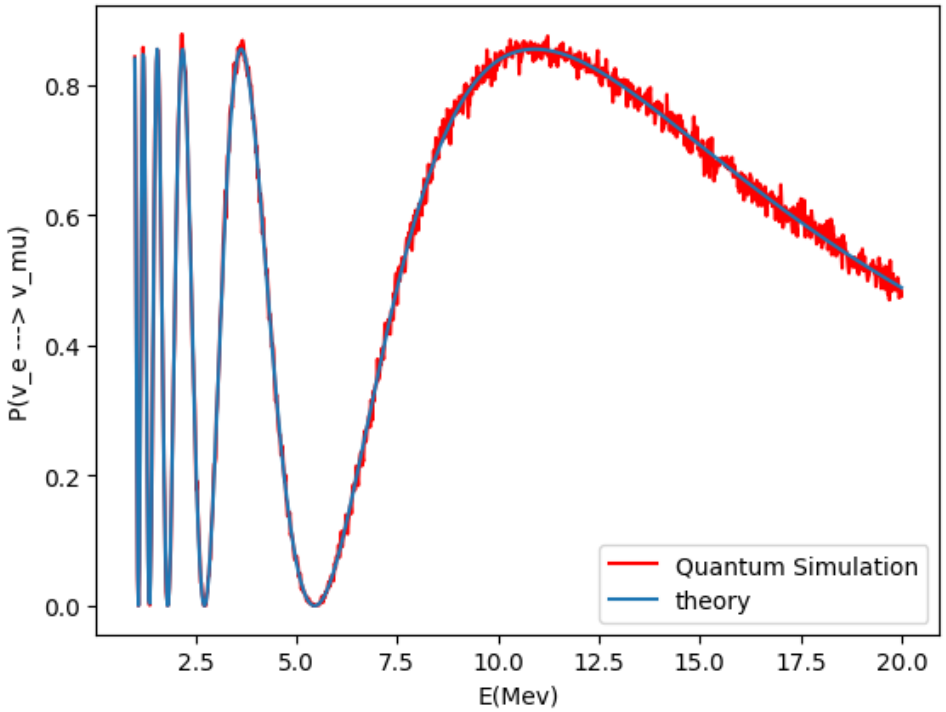


Figure 4.4: Two-flavor $\nu_e \to \nu_\mu$ appearance probability.

4.6 A simple self consistency check

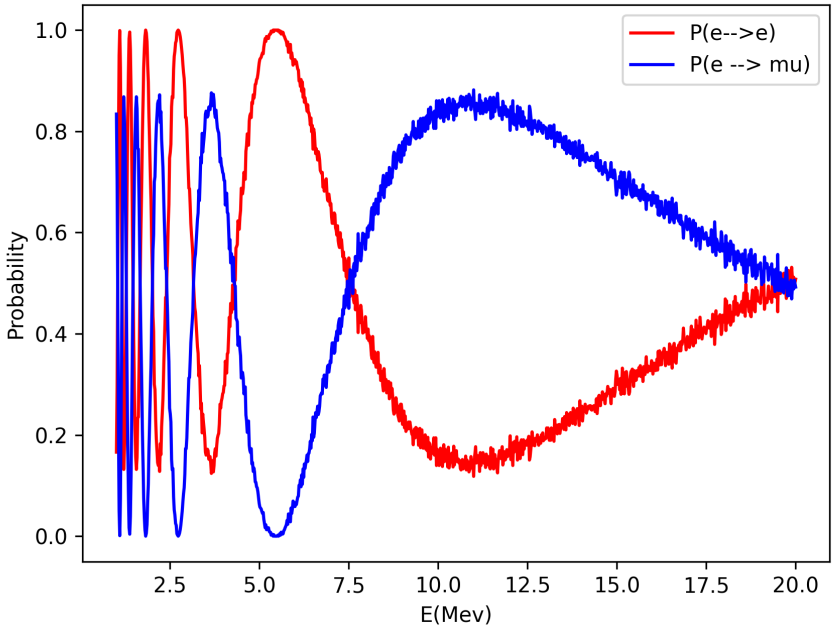


Figure 4.5: Probability Equate to 1 at every point.

Chapter 5

Three-Flavor Oscillation on a Quantum Processor

5.1 Embedding in Two Qubits

Three flavor states require a 3-dimensional Hilbert space. We embed them into the two-qubit space using the mapping:

$$|00\rangle \leftrightarrow |\nu_1\rangle, \quad |01\rangle \leftrightarrow |\nu_2\rangle, \quad |10\rangle \leftrightarrow |\nu_3\rangle, \tag{5.1}$$

while $|11\rangle$ is unused (it can be used later for sterile neutrinos). The 4×4 extension of the PMNS matrix is

$$U_{4\times4} = \begin{pmatrix} U_{\text{PMNS}}^{3\times3} & 0 \\ 0 & e^{i\phi} \end{pmatrix}, \tag{5.2}$$

where the extra phase is arbitrary and does not affect physical observables because the $|11\rangle$ state is never populated when starting from a physical flavor state[2].

5.2 Entanglement via CNOT and Condition for Separability

Consider a general two-qubit separable input state:

$$|\psi_{\text{in}}\rangle = (\alpha|0\rangle + \beta|1\rangle) \otimes (\gamma|0\rangle + \delta|1\rangle). \tag{5.3}$$

Expanding this tensor product, we obtain:

$$|\psi_{\text{in}}\rangle = \alpha\gamma|00\rangle + \alpha\delta|01\rangle + \beta\gamma|10\rangle + \beta\delta|11\rangle. \tag{5.4}$$

Now, applying the CNOT gate (with the first qubit as control and second as target), the transformation is:

$$\text{CNOT}|\psi_{\text{in}}\rangle = \alpha\gamma|00\rangle + \alpha\delta|01\rangle + \beta\gamma|11\rangle + \beta\delta|10\rangle. \tag{5.5}$$

This resulting state is, in general, **entangled**, since it cannot be written as a tensor product of two single-qubit states.

5.2.1 Condition for Separability

Assume that the output state remains separable and can be written as:

$$|\psi_{\text{out}}\rangle = (\alpha'|0\rangle + \beta'|1\rangle) \otimes (\gamma'|0\rangle + \delta'|1\rangle). \tag{5.6}$$

Expanding:

$$|\psi_{\text{out}}\rangle = \alpha'\gamma'|00\rangle + \alpha'\delta'|01\rangle + \beta'\gamma'|10\rangle + \beta'\delta'|11\rangle. \tag{5.7}$$

For separability, the coefficients must satisfy the consistency condition:

$$(\text{coeff of } |00\rangle)(\text{coeff of } |11\rangle) = (\text{coeff of } |01\rangle)(\text{coeff of } |10\rangle). \tag{5.8}$$

Substituting from the CNOT output:

$$(\alpha\gamma)(\beta\delta) = (\alpha\delta)(\beta\gamma). \tag{5.9}$$

This simplifies to:

$$\gamma^2 = \delta^2. \tag{5.10}$$

Thus, the output state remains separable only under this special constraint. In all other cases, the action of the CNOT gate generates entanglement.

5.3 Role of CNOT in PMNS Implementation

If only single-qubit rotations are used, the resulting operation on a two-qubit system takes the form

$$U = U_A \otimes U_B, \tag{5.11}$$

which represents a separable transformation acting independently on each qubit. Such operations cannot generate correlations between qubits and therefore cannot produce

entanglement. [6] However, the PMNS matrix in the three-flavor case represents a general unitary transformation embedded in a higher-dimensional Hilbert space, which requires mixing between different computational basis states. This mixing cannot be achieved using tensor products of single-qubit rotations alone[6].

To reproduce the full structure of the PMNS matrix, it is necessary to include entangling gates. The Controlled-NOT (CNOT) gate provides this capability by correlating the states of the two qubits and enabling transitions between basis states that cannot be factorized[6].

5.4 Decomposition of the PMNS Matrix

The 3×3 PMNS matrix can be decomposed into a sequence of single-qubit U_3 gates and CNOTs [2, 12]. One systematic decomposition uses the fact that any $SU(4)$ matrix can be expressed as a product of two CNOTs and six U_3 gates. For the particular embedding above, the 4×4 version of U_{PMNS} (which is block-diagonal with a 3×3 block and a 1×1 block) can be decomposed as:

$$U_{\text{PMNS}} = U_3^A(\epsilon)U_3^B(\zeta)\ \text{CNOT}_{AB}\ U_3^A(\gamma)U_3^B(\delta)\ \text{CNOT}_{AB} \tag{5.12}$$

$$\times\, U_3^A(\alpha)U_3^B(\beta), \tag{5.13}$$

using six U_3 rotations and two CNOTs. The angles $\alpha, \beta, \gamma, \delta, \epsilon, \zeta$ are determined from the PMNS parameters via a numerical fitting procedure. For our simulations we used the best-fit values from [8] and precomputed the angles; the circuit thus implements the exact mixing matrix to high precision.

5.4.1 Matrix-level proof of the Hermitian conjugate in the three-flavor case

In the three-flavor construction, each layer of single-qubit gates acting on qubits A and B is represented by a tensor product. The important point is that the Hermitian conjugate of a tensor product does *not* reverse the order:

$$(A \otimes B)^\dagger = A^\dagger \otimes B^\dagger. \tag{5.14}$$

By contrast, for an ordinary product of operators, the order does reverse:

$$(ABC)^\dagger = C^\dagger B^\dagger A^\dagger. \tag{5.15}$$

To prove this explicitly, let

$$U_A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \qquad U_B = \begin{pmatrix} e & f \\ g & h \end{pmatrix}. \tag{5.16}$$

Then their tensor product is the 4×4 matrix

$$U_A \otimes U_B = \begin{pmatrix} aU_B & bU_B \\ cU_B & dU_B \end{pmatrix} = \begin{pmatrix} ae & af & be & bf \\ ag & ah & bg & bh \\ ce & cf & de & df \\ cg & ch & dg & dh \end{pmatrix}. \tag{5.17}$$

Taking the Hermitian conjugate explicitly gives

$$(U_A \otimes U_B)^\dagger = \begin{pmatrix} (ae)^* & (ag)^* & (ce)^* & (cg)^* \\ (af)^* & (ah)^* & (cf)^* & (ch)^* \\ (be)^* & (bg)^* & (de)^* & (dg)^* \\ (bf)^* & (bh)^* & (df)^* & (dh)^* \end{pmatrix}. \tag{5.18}$$

Using $(xy)^* = x^* y^*$, this becomes

$$(U_A \otimes U_B)^\dagger = \begin{pmatrix} a^*e^* & a^*g^* & c^*e^* & c^*g^* \\ a^*f^* & a^*h^* & c^*f^* & c^*h^* \\ b^*e^* & b^*g^* & d^*e^* & d^*g^* \\ b^*f^* & b^*h^* & d^*f^* & d^*h^* \end{pmatrix}. \tag{5.19}$$

Now

$$U_A^\dagger = \begin{pmatrix} a^* & c^* \\ b^* & d^* \end{pmatrix}, \qquad U_B^\dagger = \begin{pmatrix} e^* & g^* \\ f^* & h^* \end{pmatrix}, \tag{5.20}$$

so that

$$U_A^\dagger \otimes U_B^\dagger = \begin{pmatrix} a^*U_B^\dagger & c^*U_B^\dagger \\ b^*U_B^\dagger & d^*U_B^\dagger \end{pmatrix} = \begin{pmatrix} a^*e^* & a^*g^* & c^*e^* & c^*g^* \\ a^*f^* & a^*h^* & c^*f^* & c^*h^* \\ b^*e^* & b^*g^* & d^*e^* & d^*g^* \\ b^*f^* & b^*h^* & d^*f^* & d^*h^* \end{pmatrix}. \tag{5.21}$$

Therefore,

$$\boxed{(U_A \otimes U_B)^\dagger = U_A^\dagger \otimes U_B^\dagger.} \tag{5.22}$$

This proves that the Hermitian conjugate does not interchange the order of operators inside the tensor product. The reversal occurs only between successive circuit layers.

For the three-flavor PMNS construction,

$$U_{\text{PMNS}} = (U_A(\epsilon) \otimes U_B(\zeta))\ \text{CNOT}\ (U_A(\gamma) \otimes U_B(\delta))\ \text{CNOT}\ (U_A(\alpha) \otimes U_B(\beta))\,. \quad (5.23)$$

Taking the Hermitian conjugate and using

$$(ABC)^\dagger = C^\dagger B^\dagger A^\dagger, \qquad (A \otimes B)^\dagger = A^\dagger \otimes B^\dagger, \qquad \text{CNOT}^\dagger = \text{CNOT}, \quad (5.24)$$

we obtain

$$U_{\text{PMNS}}^\dagger = (U_A(\alpha) \otimes U_B(\beta))^\dagger\ \text{CNOT}\ (U_A(\gamma) \otimes U_B(\delta))^\dagger\ \text{CNOT}\ (U_A(\epsilon) \otimes U_B(\zeta))^\dagger \quad (5.25)$$

$$= \left(U_A^\dagger(\alpha) \otimes U_B^\dagger(\beta)\right)\ \text{CNOT}\ \left(U_A^\dagger(\gamma) \otimes U_B^\dagger(\delta)\right)\ \text{CNOT}\ \left(U_A^\dagger(\epsilon) \otimes U_B^\dagger(\zeta)\right). \quad (5.26)$$

If the one-qubit gate satisfies

$$U_3^\dagger(\theta) = U_3(-\theta), \quad (5.27)$$

then the inverse PMNS circuit is

$$U_{\text{PMNS}}^\dagger = U_A(-\alpha)U_B(-\beta)\,\text{CNOT}\,U_A(-\gamma)U_B(-\delta)\,\text{CNOT}\,U_A(-\epsilon)U_B(-\zeta). \quad (5.28)$$

Thus, the full circuit order reverses, but the order inside each tensor-product layer does not.

5.5 Time Evolution Factorization

In the mass basis, the evolution operator is diagonal:

$$U_{\text{prop}} = \text{diag}\left(e^{-i\frac{m_1^2 L}{2E}}, e^{-i\frac{m_2^2 L}{2E}}, e^{-i\frac{m_3^2 L}{2E}}, 1\right). \quad (5.29)$$

Factoring out the common phase from m_1^2,

5.5.1 Mass-Basis Hamiltonian and Removal of Global Phase

In the mass basis, the effective Hamiltonian for three-flavor neutrino oscillations is

$$H = \frac{1}{2E}\begin{pmatrix} m_1^2 & 0 & 0 \\ 0 & m_2^2 & 0 \\ 0 & 0 & m_3^2 \end{pmatrix}. \quad (5.30)$$

We can rewrite this Hamiltonian by adding and subtracting m_1^2:

$$H = \frac{m_1^2}{2E} I + \frac{1}{2E} \begin{pmatrix} 0 & 0 & 0 \\ 0 & m_2^2 - m_1^2 & 0 \\ 0 & 0 & m_3^2 - m_1^2 \end{pmatrix}. \tag{5.31}$$

Defining the mass-squared differences

$$\Delta m_{21}^2 = m_2^2 - m_1^2, \qquad \Delta m_{31}^2 = m_3^2 - m_1^2, \tag{5.32}$$

the Hamiltonian becomes

$$H = \frac{m_1^2}{2E} I + \frac{1}{2E} \begin{pmatrix} 0 & 0 & 0 \\ 0 & \Delta m_{21}^2 & 0 \\ 0 & 0 & \Delta m_{31}^2 \end{pmatrix}. \tag{5.33}$$

Expanding the identity term explicitly,

$$\frac{m_1^2}{2E} I = \frac{1}{2E} \begin{pmatrix} m_1^2 & 0 & 0 \\ 0 & m_1^2 & 0 \\ 0 & 0 & m_1^2 \end{pmatrix}, \tag{5.34}$$

we verify that

$$H = \frac{1}{2E} \begin{pmatrix} m_1^2 & 0 & 0 \\ 0 & m_2^2 & 0 \\ 0 & 0 & m_3^2 \end{pmatrix}. \tag{5.35}$$

The first term, proportional to the identity, contributes only a global phase to the time-evolution operator:

$$U(t) = e^{-iHt} = e^{-i\frac{m_1^2}{2E}t} e^{-iH't}, \tag{5.36}$$

where

$$H' = \frac{1}{2E} \begin{pmatrix} 0 & 0 & 0 \\ 0 & \Delta m_{21}^2 & 0 \\ 0 & 0 & \Delta m_{31}^2 \end{pmatrix}. \tag{5.37}$$

Since a global phase does not affect physical observables, we can work with the reduced Hamiltonian H' for neutrino oscillation calculations. So the reduced mass-basis Hamiltonian,

$$H' = \frac{1}{2E} \begin{pmatrix} 0 & 0 & 0 \\ 0 & \Delta m_{21}^2 & 0 \\ 0 & 0 & \Delta m_{31}^2 \end{pmatrix}, \tag{5.38}$$

5.5.2 Time-Evolution Operator in the Mass Basis

The unitary time-evolution operator is

$$U(t) = e^{-iH't}. \tag{5.39}$$

Since the reduced Hamiltonian is diagonal,

$$H' = \begin{pmatrix} a_1 & 0 & 0 \\ 0 & a_2 & 0 \\ 0 & 0 & a_3 \end{pmatrix}, \tag{5.40}$$

its exponential can be evaluated using the series definition

$$e^{-iH't} = \sum_{n=0}^{\infty} \frac{(-itH')^n}{n!}. \tag{5.41}$$

For a diagonal matrix, each power remains diagonal:

$$(H')^n = \begin{pmatrix} a_1^n & 0 & 0 \\ 0 & a_2^n & 0 \\ 0 & 0 & a_3^n \end{pmatrix}. \tag{5.42}$$

Therefore,

$$e^{-iH't} = \sum_{n=0}^{\infty} \frac{(-it)^n}{n!} \begin{pmatrix} a_1^n & 0 & 0 \\ 0 & a_2^n & 0 \\ 0 & 0 & a_3^n \end{pmatrix}. \tag{5.43}$$

Summing term by term gives

$$e^{-iH't} = \begin{pmatrix} \sum\limits_{n=0}^{\infty} \frac{(-ita_1)^n}{n!} & 0 & 0 \\ 0 & \sum\limits_{n=0}^{\infty} \frac{(-ita_2)^n}{n!} & 0 \\ 0 & 0 & \sum\limits_{n=0}^{\infty} \frac{(-ita_3)^n}{n!} \end{pmatrix} = \begin{pmatrix} e^{-ia_1t} & 0 & 0 \\ 0 & e^{-ia_2t} & 0 \\ 0 & 0 & e^{-ia_3t} \end{pmatrix}. \tag{5.44}$$

For the three-flavor neutrino Hamiltonian,

$$H' = \frac{1}{2E} \begin{pmatrix} 0 & 0 & 0 \\ 0 & \Delta m_{21}^2 & 0 \\ 0 & 0 & \Delta m_{31}^2 \end{pmatrix}, \tag{5.45}$$

so that

$$a_1 = 0, \qquad a_2 = \frac{\Delta m^2_{21}}{2E}, \qquad a_3 = \frac{\Delta m^2_{31}}{2E}. \tag{5.46}$$

Hence the time-evolution operator is

$$U(t) = e^{-iH't} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & e^{-i\frac{\Delta m^2_{21}}{2E}t} & 0 \\ 0 & 0 & e^{-i\frac{\Delta m^2_{31}}{2E}t} \end{pmatrix}. \tag{5.47}$$

To implement this on a two-qubit quantum processor, the three-flavor Hilbert space is embedded into a four-dimensional computational basis:

$$|00\rangle,\ |01\rangle,\ |10\rangle,\ |11\rangle. \tag{5.48}$$

The evolution operator is then written as

$$U(t) = \mathrm{diag}\left(1,\ e^{-i\frac{\Delta m^2_{21}}{2E}t},\ e^{-i\frac{\Delta m^2_{31}}{2E}t},\ e^{-i\phi'_4 t}\right), \tag{5.49}$$

where the fourth phase is arbitrary because the state $|11\rangle$ is unphysical in the three-flavor case.

5.6 Choice of Unphysical Phase and Absence of Entanglement in Time Evolution

The time-evolution operator in the mass basis is diagonal and takes the form:

$$U(t) = \mathrm{diag}\left(1,\ e^{-i\frac{\Delta m^2_{21}t}{2E}},\ e^{-i\frac{\Delta m^2_{31}t}{2E}},\ e^{-i\phi'_4 t}\right). \tag{5.50}$$

Our goal is to express this operator as a tensor product of single-qubit phase gates:

$$U(t) = S_A(\phi_A) \otimes S_B(\phi_B), \tag{5.51}$$

which has the structure:

$$S_A \otimes S_B = \mathrm{diag}\left(1,\ e^{i\phi_B},\ e^{i\phi_A},\ e^{i(\phi_A+\phi_B)}\right). \tag{5.52}$$

Matching the physical phases, we choose:

$$\phi_A = -\frac{\Delta m^2_{31} t}{2E}, \qquad \phi_B = -\frac{\Delta m^2_{21} t}{2E}. \tag{5.53}$$

This fixes the phase of the fourth (unphysical) state to be:

$$\phi'_4 = \frac{\Delta m^2_{31} + \Delta m^2_{21}}{2E}. \tag{5.54}$$

With this choice, the time-evolution operator factorizes exactly as:

$$U(t) = S_A(\phi_A) \otimes S_B(\phi_B). \tag{5.55}$$

Since tensor product operations act independently on each qubit, they do not generate entanglement. Therefore, by appropriately choosing the phase of the unphysical fourth state, we ensure that the time-evolution operator remains separable and does not introduce any entangling operations. Finally, we have

$$S_A(\phi_A) \otimes S_B(\phi_B) = \mathrm{diag}\left(1,\, e^{-i\frac{\Delta m^2_{21}}{2E}t},\, e^{-i\frac{\Delta m^2_{31}}{2E}t},\, e^{-i\frac{\Delta m^2_{31}+\Delta m^2_{21}}{2E}t}\right). \tag{5.56}$$

5.7 Equivalence of the $|00\rangle$ measurement and the electron-neutrino survival probability

In the two-qubit embedding of the three-flavor system, we identify the computational basis states as

$$|00\rangle \equiv |\nu_1\rangle, \qquad |01\rangle \equiv |\nu_2\rangle, \qquad |10\rangle \equiv |\nu_3\rangle, \qquad |11\rangle \equiv |\nu_X\rangle, \tag{5.57}$$

where $|\nu_X\rangle$ is an auxiliary unphysical state. The embedded PMNS matrix is then

$$\widetilde{U} = \begin{pmatrix} U_{e1} & U_{e2} & U_{e3} & 0 \\ U_{\mu 1} & U_{\mu 2} & U_{\mu 3} & 0 \\ U_{\tau 1} & U_{\tau 2} & U_{\tau 3} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \tag{5.58}$$

The diagonal time-evolution operator in the mass basis is

$$D = \mathrm{diag}\left(e^{-i\phi_1}, e^{-i\phi_2}, e^{-i\phi_3}, e^{-i\phi_X}\right), \qquad \phi_i = \frac{m_i^2 L}{2E}. \tag{5.59}$$

Starting from the computational basis state $|00\rangle$, the output of the circuit is

$$|\psi_{\text{out}}\rangle = \widetilde{U}\, D\, \widetilde{U}^{\dagger}|00\rangle. \tag{5.60}$$

We now evaluate this expression step by step. Since

$$|00\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \tag{5.61}$$

we first obtain

$$\widetilde{U}^{\dagger}|00\rangle = \begin{pmatrix} U_{e1}^{*} \\ U_{e2}^{*} \\ U_{e3}^{*} \\ 0 \end{pmatrix} = U_{e1}^{*}|00\rangle + U_{e2}^{*}|01\rangle + U_{e3}^{*}|10\rangle. \tag{5.62}$$

Thus, $\widetilde{U}^{\dagger}$ prepares the electron-flavor state in the mass basis.

After propagation,

$$D\,\widetilde{U}^{\dagger}|00\rangle = U_{e1}^{*}e^{-i\phi_1}|00\rangle + U_{e2}^{*}e^{-i\phi_2}|01\rangle + U_{e3}^{*}e^{-i\phi_3}|10\rangle. \tag{5.63}$$

Finally, applying $\widetilde{U}$ and projecting onto $\langle 00|$, the coefficient of $|00\rangle$ in the output state is

$$c_{00} = \langle 00|\widetilde{U} D \widetilde{U}^{\dagger}|00\rangle. \tag{5.64}$$

Using the first row of $\widetilde{U}$, this becomes

$$c_{00} = U_{e1}e^{-i\phi_1}U_{e1}^{*} + U_{e2}e^{-i\phi_2}U_{e2}^{*} + U_{e3}e^{-i\phi_3}U_{e3}^{*} \tag{5.65}$$

$$= |U_{e1}|^2 e^{-i\phi_1} + |U_{e2}|^2 e^{-i\phi_2} + |U_{e3}|^2 e^{-i\phi_3}. \tag{5.66}$$

On the other hand, the standard three-flavor electron-neutrino survival amplitude is

$$A_{ee} = \langle \nu_e|\nu_e(L)\rangle = \sum_{i=1}^{3} U_{ei}e^{-i\phi_i}U_{ei}^{*} = \sum_{i=1}^{3}|U_{ei}|^2 e^{-i\phi_i}. \tag{5.67}$$

Therefore,

$$c_{00} = A_{ee}. \tag{5.68}$$

Taking the modulus squared of both sides immediately gives

$$|c_{00}|^2 = |A_{ee}|^2 = P(\nu_e \to \nu_e). \tag{5.69}$$

Hence, the probability of measuring the state $|00\rangle$ at the output of the quantum circuit is exactly the electron-neutrino survival probability:

$$\boxed{P(\nu_e \to \nu_e) = \left|\langle 00|\widetilde{U} D \widetilde{U}^\dagger |00\rangle\right|^2 = |c_{00}|^2}. \tag{5.70}$$

Removing the overall unphysical phase $e^{-i\phi_1}$, one may also write

$$c_{00} = |U_{e1}|^2 + |U_{e2}|^2 e^{-i\Delta_{21}} + |U_{e3}|^2 e^{-i\Delta_{31}}, \tag{5.71}$$

where

$$\Delta_{ij} = \frac{\Delta m_{ij}^2 L}{2E}. \tag{5.72}$$

Therefore,

$$P(\nu_e \to \nu_e) = \left||U_{e1}|^2 + |U_{e2}|^2 e^{-i\Delta_{21}} + |U_{e3}|^2 e^{-i\Delta_{31}}\right|^2. \tag{5.73}$$

5.8 Identification of the $|01\rangle$ and $|10\rangle$ measurement probabilities

In the two-qubit embedding of the three-flavor system, the final state for an initial electron neutrino is

$$|\psi_{\text{out}}\rangle = \widetilde{U} D \widetilde{U}^\dagger |00\rangle, \tag{5.74}$$

where

$$\widetilde{U} = \begin{pmatrix} U_{e1} & U_{e2} & U_{e3} & 0 \\ U_{\mu 1} & U_{\mu 2} & U_{\mu 3} & 0 \\ U_{\tau 1} & U_{\tau 2} & U_{\tau 3} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \tag{5.75}$$

and

$$D = \text{diag}\left(e^{-i\phi_1}, e^{-i\phi_2}, e^{-i\phi_3}, e^{-i\phi_X}\right), \qquad \phi_i = \frac{m_i^2 L}{2E}. \tag{5.76}$$

Starting from $|00\rangle$, one first obtains

$$\widetilde{U}^\dagger |00\rangle = U_{e1}^* |00\rangle + U_{e2}^* |01\rangle + U_{e3}^* |10\rangle. \tag{5.77}$$

After propagation,

$$D\widetilde{U}^\dagger |00\rangle = U_{e1}^* e^{-i\phi_1} |00\rangle + U_{e2}^* e^{-i\phi_2} |01\rangle + U_{e3}^* e^{-i\phi_3} |10\rangle. \tag{5.78}$$

The coefficient of $|01\rangle$ in the final state is

$$c_{01} = \langle 01| \widetilde{U} D \widetilde{U}^\dagger |00\rangle. \tag{5.79}$$

Using the second row of $\widetilde{U}$, this becomes

$$c_{01} = U_{\mu 1} e^{-i\phi_1} U_{e1}^* + U_{\mu 2} e^{-i\phi_2} U_{e2}^* + U_{\mu 3} e^{-i\phi_3} U_{e3}^*. \tag{5.80}$$

This is precisely the standard oscillation amplitude

$$A_{e\mu} = \sum_{i=1}^{3} U_{\mu i} e^{-i\phi_i} U_{ei}^*. \tag{5.81}$$

Therefore,

$$c_{01} = A_{e\mu}, \tag{5.82}$$

and hence

$$|c_{01}|^2 = |A_{e\mu}|^2 = P(\nu_e \to \nu_\mu). \tag{5.83}$$

Similarly, the coefficient of $|10\rangle$ is

$$c_{10} = \langle 10| \widetilde{U} D \widetilde{U}^\dagger |00\rangle. \tag{5.84}$$

Using the third row of $\widetilde{U}$, one finds

$$c_{10} = U_{\tau 1} e^{-i\phi_1} U_{e1}^* + U_{\tau 2} e^{-i\phi_2} U_{e2}^* + U_{\tau 3} e^{-i\phi_3} U_{e3}^*. \tag{5.85}$$

This is exactly the standard oscillation amplitude

$$A_{e\tau} = \sum_{i=1}^{3} U_{\tau i} e^{-i\phi_i} U_{ei}^*. \tag{5.86}$$

Therefore,

$$c_{10} = A_{e\tau}, \tag{5.87}$$

and hence

$$|c_{10}|^2 = |A_{e\tau}|^2 = P(\nu_e \to \nu_\tau). \tag{5.88}$$

Thus, the output probabilities of the three physical computational basis states are

$$|c_{00}|^2 = P(\nu_e \to \nu_e), \qquad |c_{01}|^2 = P(\nu_e \to \nu_\mu), \qquad |c_{10}|^2 = P(\nu_e \to \nu_\tau). \tag{5.89}$$

5.9 Full Circuit

The full three-flavor circuit combines the PMNS decomposition (six U_3 gates and two CNOTs)[2], the two phase gates for propagation, and the inverse PMNS decomposition (another six U_3 gates and two CNOTs). The total gate count is therefore 12 single-qubit rotations and 4 CNOTs. We implemented this circuit for various initial flavors and baselines on QISKIT. Figures 5.4–5.9 show the appearance probabilities as functions of L/E. The simulations reproduce the characteristic three-flavor interference patterns, including the effects of the small θ_{13} and the atmospheric/solar oscillations. The hardware results show slight damping due to decoherence, which is consistent with the decoherence model discussed later.

5.9.1 Three flavour neutrino oscillation experiment as run on QISKIT

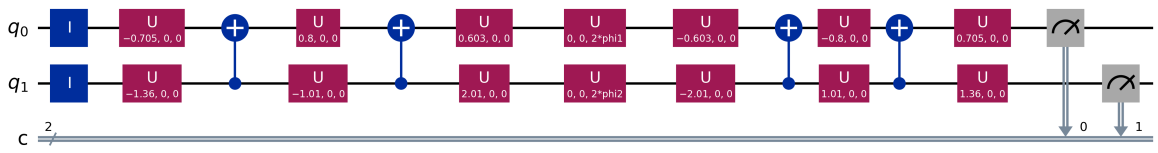


Figure 5.1: This circuit give the probability of $\nu_e \to \nu_\alpha$ appearance for small L/E, here$\alpha = e, \mu, \tau$.

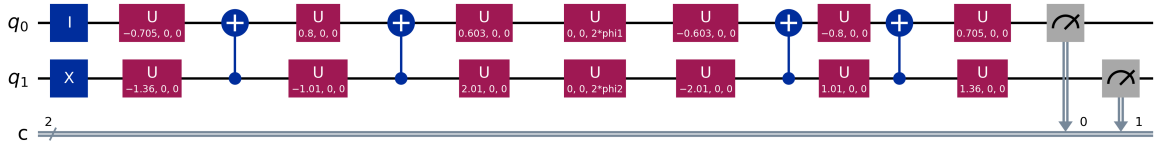


Figure 5.2: This circuit give the probability of $\nu_\mu \to \nu_\alpha$ appearance for small L/E, here$\alpha = e, \mu, \tau$.

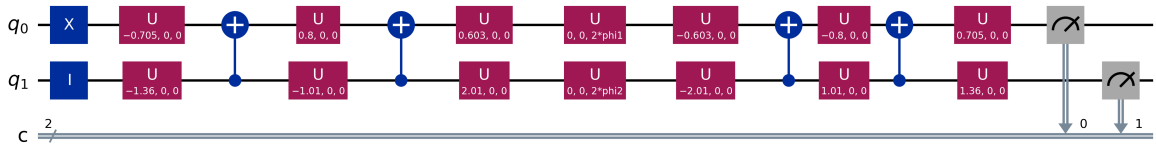


Figure 5.3: This circuit give the probability of $\nu_\tau \to \nu_\alpha$ appearance for small L/E, here$\alpha = e, \mu, \tau$.

5.10 Probability Graph and Results

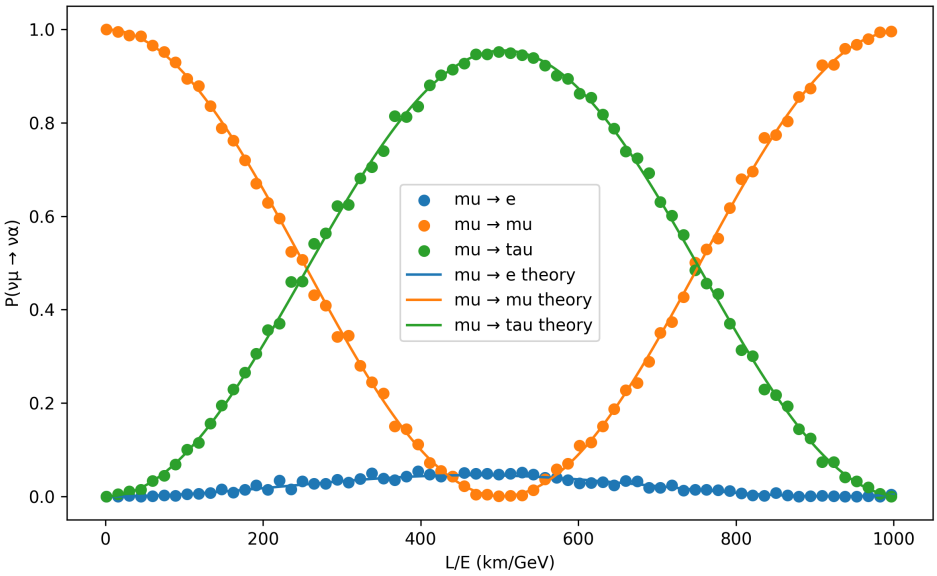


Figure 5.4: $\nu_\mu \to \nu_\alpha$ appearance for small L/E.

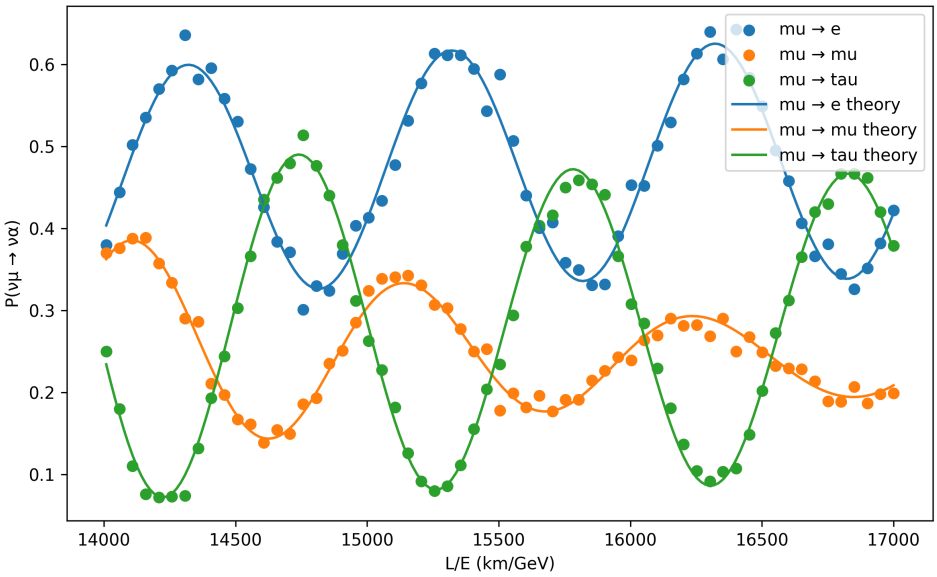


Figure 5.5: $\nu_\mu \to \nu_\alpha$ appearance for large L/E.

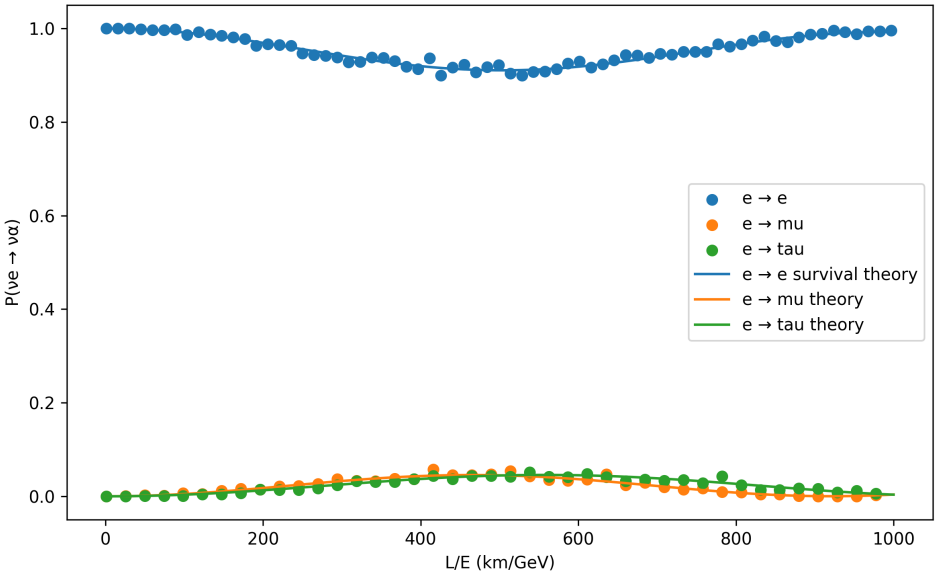


Figure 5.6: $\nu_e \to \nu_\alpha$ appearance for small L/E.

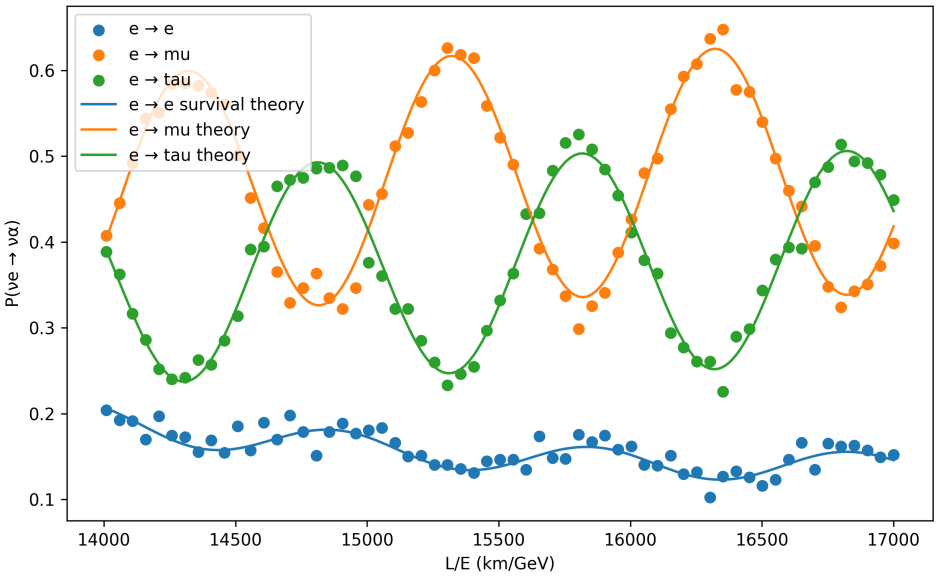


Figure 5.7: $\nu_e \to \nu_\alpha$ appearance for large L/E.

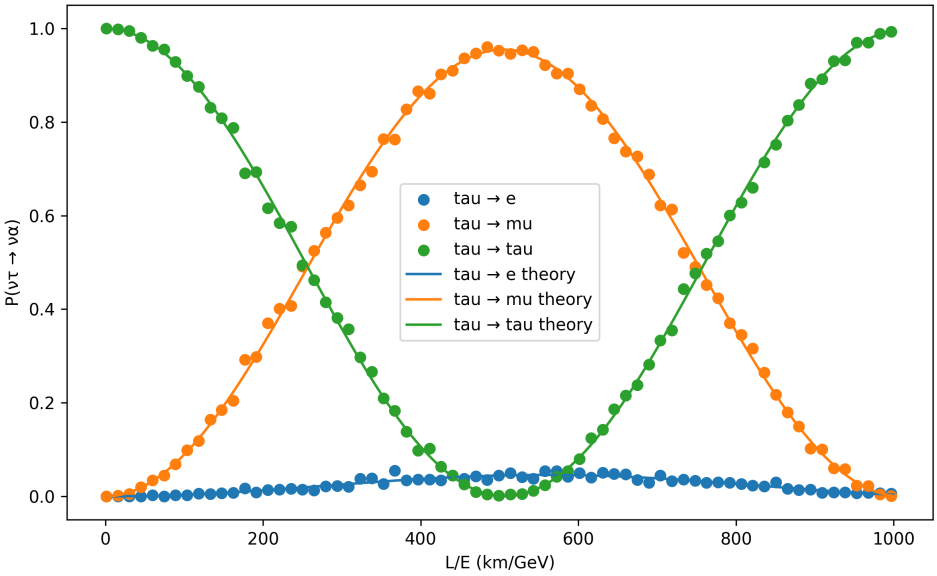


Figure 5.8: $\nu_\tau \to \nu_\alpha$ appearance for small L/E.

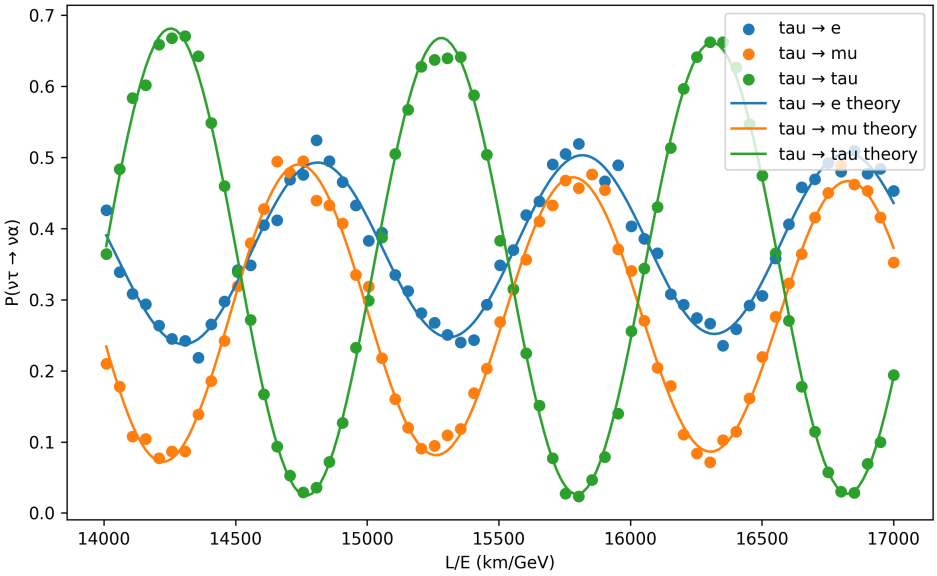


Figure 5.9: $\nu_\tau \to \nu_\alpha$ appearance for large L/E.

Chapter 6

Sterile Neutrinos

6.1 3+1 Framework

The sterile neutrino hypothesis introduces a fourth mass eigenstate ν_4 and a new flavor ν_s. The 4×4 PMNS matrix includes three additional mixing angles $\theta_{14}, \theta_{24}, \theta_{34}$ and two new CP phases. Global fits to short-baseline anomalies suggest $\Delta m^2_{41} \sim 1$ eV^2 [3]. The mixing matrix is typically parameterized as

$$U_{3+1} = R_{34} R_{24} R_{14} R_{23} R_{13} R_{12}, \tag{6.1}$$

where R_{ij} are rotation matrices in the i–j plane. In the extended 4×4 space, the flavor states are embedded as $|\nu_\alpha\rangle$ ($\alpha = e, \mu, \tau$) in the first three dimensions, and $|\nu_s\rangle$ corresponds to the fourth. The $|11\rangle$ state in the two-qubit encoding now represents $|\nu_4\rangle$ (or equivalently $|\nu_s\rangle$ after mixing), so the full 4×4 PMNS matrix acts on the two-qubit space.

6.2 Time Evolution and Entanglement

After removing a global phase, the propagation operator becomes

$$U_{\text{prop}} = \text{diag}\left(1, e^{-i\alpha}, e^{-i\beta}, e^{-i\gamma}\right), \tag{6.2}$$

with $\alpha = \frac{\Delta m^2_{21} L}{2E}$, $\beta = \frac{\Delta m^2_{31} L}{2E}$, $\gamma = \frac{\Delta m^2_{41} L}{2E}$ (up to redefinitions). This operator does not factorize as a tensor product of single-qubit gates because the fourth phase is not a sum of the first two. Consequently, propagation generates entanglement. For instance, starting from a separable state $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$, the evolved state $\frac{1}{2}(|00\rangle + e^{-i\alpha}|01\rangle + e^{-i\beta}|10\rangle + e^{-i\gamma}|11\rangle)$ is entangled unless $\alpha + \gamma = \beta$ (which is not satisfied for independent

masses). This entanglement is a resource that must be generated by the circuit[2].

6.3 Circuit Implementation

A minimal circuit that produces three independent phases uses two CNOTs and three single-qubit phase gates:

$$U_{\text{prop}} = (S_A \otimes S_B)\ \text{CNOT}\ U_1(\lambda)\ \text{CNOT}, \tag{6.3}$$

where S_A and S_B provide phases α and β, and the central CNOT–U_1–CNOT block implements a controlled-phase gate $\text{diag}(1, 1, 1, e^{i\lambda})$, providing the third phase γ.

6.3.1 Definition of the Phases α, β, γ

The diagonal unitary obtained from the circuit is written as

$$\text{diag}\Big(1,\ e^{i\alpha},\ e^{i\beta},\ e^{i\gamma}\Big).$$

In the neutrino oscillation framework, these phases arise from the propagation in the mass basis. For a four-dimensional embedding, the time-evolution operator can be written as

$$U_{\text{mass}} = \text{diag}\left(e^{-i\frac{m_1^2 L}{2E}},\ e^{-i\frac{m_2^2 L}{2E}},\ e^{-i\frac{m_3^2 L}{2E}},\ e^{-i\frac{m_4^2 L}{2E}}\right).$$

Factoring out a global phase $e^{-i\frac{m_1^2 L}{2E}}$, we obtain

$$U_{\text{mass}} \sim \text{diag}\left(1,\ e^{-i\frac{\Delta m_{21}^2 L}{2E}},\ e^{-i\frac{\Delta m_{31}^2 L}{2E}},\ e^{-i\frac{\Delta m_{41}^2 L}{2E}}\right),$$

where $\Delta m_{ij}^2 = m_i^2 - m_j^2$.

Thus, α, β, and γ represent the relative dynamical phases accumulated due to the mass-squared differences with respect to the reference state m_1.

6.4 Phase Matching in the Two-Qubit Decomposition

We consider the circuit

$$\Big(S(\phi_1) \otimes S(\phi_2)\Big)\ \text{CNOT}\ (I \otimes S(\phi_3))\ \text{CNOT},$$

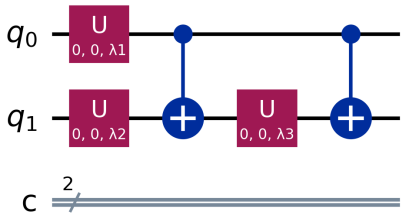


Figure 6.1: Time evolution circuit for neutrino oscillation

which produces a diagonal two-qubit unitary of the form

$$\mathrm{diag}\left(1,\ e^{i\alpha},\ e^{i\beta},\ e^{i\gamma}\right).$$

Matching the diagonal entries gives the relations

$$\alpha = \phi_2 + \phi_3, \tag{6.4}$$
$$\beta = \phi_1 + \phi_3, \tag{6.5}$$
$$\gamma = \phi_1 + \phi_2. \tag{6.6}$$

Solving for the phases ϕ_i, we obtain

$$\phi_1 = \frac{\beta + \gamma - \alpha}{2}, \tag{6.7}$$
$$\phi_2 = \frac{\alpha + \gamma - \beta}{2}, \tag{6.8}$$
$$\phi_3 = \frac{\alpha + \beta - \gamma}{2}. \tag{6.9}$$

Chapter 7

Decoherence

7.1 Lindblad Formalism

Realistic neutrino propagation may involve decoherence due to interactions with the environment (e.g., wave-packet effects, scattering). We model this via the Lindblad master equation[13, 4, 2]:

$$\frac{d\rho}{dt} = -i[H, \rho] - \mathcal{D}[\rho], \tag{7.1}$$

where $\mathcal{D}$ describes pure dephasing in the mass basis:

$$\frac{d\rho_{ij}}{dt} = -i(E_i - E_j)\rho_{ij} - \gamma_{ij}\rho_{ij} \quad (i \neq j), \qquad \frac{d\rho_{ii}}{dt} = 0. \tag{7.2}$$

The solution is

$$\rho_{ij}(L) = \rho_{ij}(0)\, e^{-i\frac{\Delta m^2_{ij} L}{2E}}\, e^{-\gamma_{ij} L}. \tag{7.3}$$

Thus decoherence introduces an exponential damping factor on off-diagonal elements. For simplicity we assume a single decoherence rate γ for all off-diagonals, which is typical of isotropic environments.

7.2 Quantum Circuit for Decoherence

We simulate decoherence by coupling the system to an ancilla qubit. The ancilla is prepared in $|0\rangle$, and we apply a conditional rotation that entangles it with the system[2]:

$$|i\rangle|0\rangle \to |i\rangle(\cos\theta_i|0\rangle + \sin\theta_i|1\rangle). \tag{7.4}$$

Tracing out the ancilla yields the damping factor $\cos(\theta_i - \theta_j)$ on the off-diagonal elements. By choosing θ_i such that $\cos(\theta_i - \theta_j) = e^{-\gamma_{ij} L}$, we can implement the desired damping.

For two levels, setting $\theta_1 = 0$, $\theta_2 = \theta$ gives $\cos\theta = e^{-\gamma L}$, so $\theta = \arccos(e^{-\gamma L})$.

A circuit that achieves this for a two-level system (mass basis) is:

$$\text{CNOT } U_3(\theta, 0, 0) \text{ CNOT } U_3(-\theta, 0, 0), \tag{7.5}$$

where the CNOTs use the system as control and the ancilla as target. The U_3 rotations

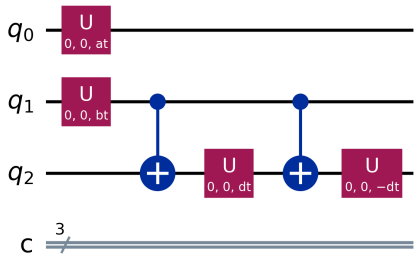


Figure 7.1: Time evolution circuit for anomalous decoherence in the mass basis

are applied to the ancilla. This block is inserted between the propagation and the final mixing. For a multi-level system, a series of such blocks can be used, each targeting a different pair of levels. The ancilla is measured at the end and the results are discarded, effectively implementing the partial trace.

Chapter 8

Matter Effects

8.1 Matter Effects in Two-Flavor Neutrino Oscillations

In vacuum, two-flavor neutrino oscillations are governed by the Hamiltonian [9, 10, 1]

$$H_{\text{vac}} = \frac{\Delta m^2}{4E} \begin{pmatrix} -\cos 2\theta & \sin 2\theta \\ \sin 2\theta & \cos 2\theta \end{pmatrix}, \tag{8.1}$$

where $\Delta m^2 = m_2^2 - m_1^2$, E is the neutrino energy, and θ is the vacuum mixing angle.

When neutrinos propagate through matter, the electron neutrino experiences an additional effective potential due to coherent forward scattering. This matter contribution is written as

$$V_{\text{mat}} = \begin{pmatrix} V & 0 \\ 0 & 0 \end{pmatrix}, \tag{8.2}$$

where V denotes the matter potential. The total Hamiltonian in the flavor basis is therefore

$$H = H_{\text{vac}} + V_{\text{mat}} = \begin{pmatrix} V - \dfrac{\Delta m^2}{4E}\cos 2\theta & \dfrac{\Delta m^2}{4E}\sin 2\theta \\ \dfrac{\Delta m^2}{4E}\sin 2\theta & \dfrac{\Delta m^2}{4E}\cos 2\theta \end{pmatrix}. \tag{8.3}$$

For convenience, define

$$A = \frac{\Delta m^2}{2E}. \tag{8.4}$$

Then the Hamiltonian may be written in the compact form

$$H = \begin{pmatrix} V - \dfrac{A}{2}\cos 2\theta & \dfrac{A}{2}\sin 2\theta \\ \dfrac{A}{2}\sin 2\theta & \dfrac{A}{2}\cos 2\theta \end{pmatrix}. \tag{8.5}$$

Let

$$H = \begin{pmatrix} a & b \\ b & d \end{pmatrix}. \tag{8.6}$$

The eigenvalues of this symmetric Hamiltonian are

$$\lambda_{\pm} = \frac{a+d}{2} \pm \sqrt{\left(\frac{a-d}{2}\right)^2 + b^2}. \tag{8.7}$$

Substituting the matrix elements gives

$$\lambda_{\pm} = \frac{V}{2} \pm \frac{1}{2}\sqrt{(V - A\cos 2\theta)^2 + (A\sin 2\theta)^2}. \tag{8.8}$$

This motivates the definition of the effective mass-squared splitting in matter,

$$\Delta m^2_m = \sqrt{(\Delta m^2 \cos 2\theta - 2EV)^2 + (\Delta m^2 \sin 2\theta)^2}. \tag{8.9}$$

Using this, the eigenvalues can be written as

$$\lambda_{\pm} = \frac{V}{2} \pm \frac{\Delta m^2_m}{4E}. \tag{8.10}$$

The time-evolution operator in the instantaneous matter eigenbasis is

$$U(t) = \begin{pmatrix} e^{-i\lambda_- t} & 0 \\ 0 & e^{-i\lambda_+ t} \end{pmatrix}. \tag{8.11}$$

Factoring out the common phase,

$$U(t) = e^{-iVt/2} \begin{pmatrix} e^{+i\Delta m^2_m t/(4E)} & 0 \\ 0 & e^{-i\Delta m^2_m t/(4E)} \end{pmatrix}. \tag{8.12}$$

The overall phase factor $e^{-iVt/2}$ is physically irrelevant, since only the eigenvalue difference contributes to oscillation probabilities. Thus, the oscillation frequency is controlled by

$$\Delta\lambda = \frac{\Delta m^2_m}{2E}. \tag{8.13}$$

The Hamiltonian in matter can be diagonalized by a rotation with an effective mixing angle θ_m,

$$R(\theta_m) = \begin{pmatrix} \cos\theta_m & \sin\theta_m \\ -\sin\theta_m & \cos\theta_m \end{pmatrix}, \tag{8.14}$$

such that

$$R^T(\theta_m)\, H\, R(\theta_m) = \text{diagonal}. \tag{8.15}$$

Requiring the off-diagonal term to vanish gives

$$\frac{1}{2}(d-a)\sin 2\theta_m + b\cos 2\theta_m = 0, \tag{8.16}$$

from which one obtains

$$\tan 2\theta_m = \frac{2b}{d-a} = \frac{\Delta m^2 \sin 2\theta}{\Delta m^2 \cos 2\theta - 2EV}. \tag{8.17}$$

Finally, the full evolution operator in the flavor basis is

$$U_f(t) = R(\theta_m) \begin{pmatrix} e^{-i\lambda_- t} & 0 \\ 0 & e^{-i\lambda_+ t} \end{pmatrix} R^\dagger(\theta_m). \tag{8.18}$$

This has the same structure as the vacuum evolution operator, with the replacements

$$\theta \to \theta_m, \qquad \Delta m^2 \to \Delta m^2_m. \tag{8.19}$$

Therefore, matter modifies both the effective mixing angle and the oscillation frequency of the neutrino system.

8.2 Implementation on a Quantum Processor

[2]Because the structure is identical to the vacuum case, the same circuit can be used, but with the mixing angle and oscillation frequency replaced by their matter-modified values. The only additional input is the matter potential V, which is a constant given the density. We have implemented this circuit for various densities and verified that the oscillation probabilities change as expected. Notably, for a fixed L/E, the probability becomes a function of θ_m and Δm^2_m, which can be precomputed classically and then used to set the gate angles. This approach works for both two-flavor and three-flavor cases, though in three flavors the matter Hamiltonian is not diagonal in the same basis and requires a more involved treatment (e.g., using the exact diagonalization to obtain the effective mixing matrix in matter).

Chapter 9

CP Violation in Neutrino Oscillations

Charge–parity (CP) violation in the lepton sector is one of the most important open questions in particle physics. A nonzero CP-violating phase in the leptonic mixing matrix would imply that neutrinos and antineutrinos oscillate with different probabilities, a phenomenon of direct relevance to leptogenesis scenarios that attempt to explain the observed matter–antimatter asymmetry of the Universe. Long-baseline experiments such as T2K, NOνA, DUNE, and Hyper-Kamiokande are designed to measure this asymmetry. In this section we extend the quantum-circuit framework developed in the previous sections to include the Dirac CP phase δ_{CP} of the PMNS matrix, and we verify that the resulting circuit reproduces the CP-violating oscillation probabilities.

9.1 The CP Phase in the PMNS Matrix

In the standard parametrization, the three-flavor PMNS matrix is a product of three rotations, with the complex phase δ_{CP} attached to the 1–3 sector:

$$U_{\mathrm{PMNS}} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & c_{23} & s_{23} \\ 0 & -s_{23} & c_{23} \end{pmatrix} \begin{pmatrix} c_{13} & 0 & s_{13}e^{-i\delta_{CP}} \\ 0 & 1 & 0 \\ -s_{13}e^{i\delta_{CP}} & 0 & c_{13} \end{pmatrix} \begin{pmatrix} c_{12} & s_{12} & 0 \\ -s_{12} & c_{12} & 0 \\ 0 & 0 & 1 \end{pmatrix}, \tag{9.1}$$

where $c_{ij} = \cos\theta_{ij}$ and $s_{ij} = \sin\theta_{ij}$. For $\delta_{CP} \neq 0, \pi$ the matrix is genuinely complex, and this complexity cannot be removed by rephasing the lepton fields. It is this irreducible phase that generates CP violation in oscillations. In the two-flavor limit the mixing matrix is real, which is why CP violation is intrinsically a three-flavor effect.

9.2 How CP Violation Appears in the Oscillation Probability

Starting from the amplitude $\mathcal{A}(\nu_\alpha \to \nu_\beta) = \sum_i U_{\beta i}\, e^{-i m_i^2 L/2E}\, U^*_{\alpha i}$ and squaring, the general three-flavor oscillation probability can be written as

$$P_{\alpha\to\beta} = \delta_{\alpha\beta} - 4\sum_{i<j} \Re\!\left(U^*_{\alpha i} U_{\beta i} U_{\alpha j} U^*_{\beta j}\right) \sin^2\!\left(\frac{\Delta m^2_{ij} L}{4E}\right) + 2\sum_{i<j} \Im\!\left(U^*_{\alpha i} U_{\beta i} U_{\alpha j} U^*_{\beta j}\right) \sin\!\left(\frac{\Delta m^2_{ij} L}{2E}\right). \tag{9.2}$$

The structure of Eq. (9.2) makes the role of the CP phase transparent:

- The first oscillatory term, proportional to $\sin^2\!\left(\Delta m^2_{ij} L/4E\right)$, involves only the real part of the quartic product of mixing-matrix elements and is *CP-even*: it is identical for neutrinos and antineutrinos.
- The second oscillatory term, proportional to $\sin\!\left(\Delta m^2_{ij} L/2E\right)$, involves the imaginary part of the quartic product and is *CP-odd*: it changes sign under CP conjugation.
- CP violation therefore occurs if and only if $\Im\!\left(U^*_{\alpha i} U_{\beta i} U_{\alpha j} U^*_{\beta j}\right) \neq 0$ for $\alpha \neq \beta$.

For three flavors, unitarity forces all such imaginary parts to be proportional to a single rephasing-invariant quantity, the Jarlskog invariant,

$$J = \Im\!\left(U^*_{e1} U_{\mu 1} U_{e2} U^*_{\mu 2}\right) = s_{12} c_{12}\, s_{23} c_{23}\, s_{13} c_{13}^2\, \sin\delta_{CP}, \tag{9.3}$$

so that the CP-odd term in every appearance channel is controlled by $J \propto \sin\delta_{CP}$. Note also that the CP-odd term vanishes identically in survival channels ($\alpha = \beta$), since the quartic product is then real; CP violation is observable only in appearance channels.

9.3 Neutrino–Antineutrino Difference

Under CP conjugation, a neutrino oscillation $\nu_\alpha \to \nu_\beta$ is mapped to the antineutrino oscillation $\bar\nu_\alpha \to \bar\nu_\beta$. At the amplitude level this is implemented by complex-conjugating the mixing matrix:

$$U_{\text{PMNS}} \;\longrightarrow\; U^*_{\text{PMNS}}, \tag{9.4}$$

equivalently $\delta_{CP} \to -\delta_{CP}$. The real parts in Eq. (9.2) are unchanged, while the imaginary parts flip sign, giving

$$P(\bar{\nu}_\alpha \to \bar{\nu}_\beta) = \delta_{\alpha\beta} - 4\sum_{i<j} \Re\left(U^*_{\alpha i} U_{\beta i} U_{\alpha j} U^*_{\beta j}\right) \sin^2\left(\frac{\Delta m^2_{ij} L}{4E}\right) - 2\sum_{i<j} \Im\left(U^*_{\alpha i} U_{\beta i} U_{\alpha j} U^*_{\beta j}\right) \sin\left(\frac{\Delta m^2_{ij} L}{2E}\right). \tag{9.5}$$

Subtracting Eq. (9.5) from Eq. (9.2), the CP asymmetry is

$$\Delta P^{CP}_{\alpha\beta} \equiv P(\nu_\alpha \to \nu_\beta) - P(\bar{\nu}_\alpha \to \bar{\nu}_\beta) = 4\sum_{i<j} \Im\left(U^*_{\alpha i} U_{\beta i} U_{\alpha j} U^*_{\beta j}\right) \sin\left(\frac{\Delta m^2_{ij} L}{2E}\right) \neq 0 \tag{9.6}$$

whenever the imaginary part is nonzero, i.e. whenever $\sin\delta_{CP} \neq 0$ (and all mixing angles are nonvanishing). This asymmetry is the quantity that experiments target, and it is also a convenient diagnostic for our circuit: simulating the same channel with U_{PMNS} and U^*_{PMNS} must produce probability curves that differ by exactly Eq. (9.6).

An important practical remark is that on the quantum circuit, the antineutrino case requires no new construction: since conjugating U_{PMNS} amounts to $\delta_{CP} \to -\delta_{CP}$, the same circuit with the sign of every CP-dependent phase parameter reversed simulates antineutrino propagation.

9.4 Circuit Strategy for the CP-Violating Case

In the CP-conserving case discussed earlier, the 3×3 PMNS matrix was real, and its 4×4 embedding

$$U_{4\times4} = \begin{pmatrix} U^{\text{PMNS}}_{3\times3} & 0 \\ 0 & e^{i\phi} \end{pmatrix} \tag{9.7}$$

could be synthesized from six real single-qubit rotations $U_3(\theta, 0, 0)$ and two CNOT gates. With $\delta_{CP} \neq 0$ the target matrix is complex, and rotations about a single axis are no longer sufficient: the circuit must be able to generate relative phases between computational basis states in addition to real mixing.

To supply these phases we augment the decomposition with single-qubit phase gates,

$$S(\phi) = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\phi} \end{pmatrix}, \tag{9.8}$$

applied to qubits A and B. A layer $S_A(\phi_A) \otimes S_B(\phi_B)$ acts on the two-qubit space as $\text{diag}\left(1,\, e^{i\phi_B},\, e^{i\phi_A},\, e^{i(\phi_A+\phi_B)}\right)$, which is exactly the freedom needed to place the phase $e^{\pm i\delta_{CP}}$ on the correct elements of the embedded PMNS matrix while leaving the redundant 11 state decoupled from the physical three-dimensional subspace. The PMNS gate is then constructed as shown in Fig. 9.1.

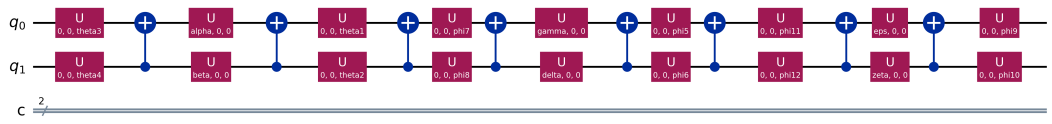


Figure 9.1: PMNS gate with CP violation. The phase gates $S_A(\phi)$ and $S_B(\phi)$ are introduced to handle the imaginary part coming from the CP phase in U_{PMNS}, while the U_3 rotations and CNOT gates provide the real mixing structure as in the CP-conserving construction.

The essential points of the construction are:

- the phase gates $S_A(\phi)$ and $S_B(\phi)$ were introduced to handle the imaginary part coming from the CP phase in U_{PMNS};
- the $U_3(\theta, 0, 0)$ rotations and the two CNOT gates retain the same roles as in the CP-conserving decomposition, generating the real orthogonal part of the mixing;
- the mass-basis time-evolution block is unchanged, since the Hamiltonian in the mass basis remains diagonal and real; all CP dependence enters exclusively through the mixing gates $U^{\dagger}_{\text{PMNS}}$ and U_{PMNS}.

The full two-qubit simulation circuit retains the structure

$$U^{\dagger}_{\text{PMNS}}(\delta_{CP}) \;\rightarrow\; U(t) \;\rightarrow\; U_{\text{PMNS}}(\delta_{CP}), \tag{9.9}$$

i.e. rotation from the flavor basis to the mass basis, free propagation as a tensor product of phase gates, and rotation back to the flavor basis, followed by measurement of both qubits in the computational basis. The circuit used for the electron-neutrino appearance channel is shown in Fig. 9.2.

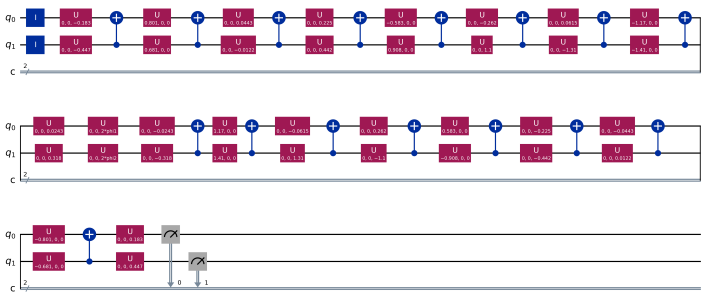


Figure 9.2: Two-qubit quantum circuit implementing CP-violating three-flavor oscillations, shown here for the $\nu_e \rightarrow \nu_\alpha$ appearance channel.

9.4.1 Matching the Circuit to U_{PMNS} with a CP Phase

As in the CP-conserving case, each layer of single-qubit gates contributes a 4×4 tensor-product unitary, and multiplying all layers together with the CNOT gates yields a single effective two-qubit unitary U_{eff} depending on the gate parameters $(\alpha, \beta, \gamma, \delta, \epsilon, \zeta, \phi_A, \phi_B, \dots)$. The gate parameters are determined by demanding

$$U_{\mathrm{eff}} \;\approx\; U_{\mathrm{PMNS}}^{(4\times4)}(\delta_{CP}), \tag{9.10}$$

where $U_{\mathrm{PMNS}}^{(4\times4)}(\delta_{CP})$ is the block embedding of the complex PMNS matrix of Eq. (9.1). In practice the matching is performed numerically, minimizing a cost function such as

$$C = 1 - \frac{1}{4}\left|\mathrm{Tr}\left(U_{\mathrm{eff}}^{\dagger}\, U_{\mathrm{PMNS}}^{(4\times4)}(\delta_{CP})\right)\right|, \tag{9.11}$$

which vanishes when the two unitaries agree up to an irrelevant global phase. Because the physical dynamics lives entirely in the three-dimensional flavor subspace, only the upper-left 3×3 block must be matched exactly; the phase $e^{i\phi}$ associated with the redundant 11 state is left free, which enlarges the solution space of the fit. The fitted parameters reduce smoothly to the real, CP-conserving decomposition when $\delta_{CP} \to 0$, providing a consistency check on the construction.

9.5 Results: Theory versus Quantum Circuit

To validate the construction, the CP-incorporated circuit was run for the $\nu_e \to \nu_\alpha$ channels and compared against the analytic probabilities of Eq. (9.2), using best-fit values of the mixing angles, mass-squared splittings, and a nonzero δ_{CP}. Figures 9.3 and 9.4 show the resulting oscillation probabilities in the small and large L/E regimes, respectively.

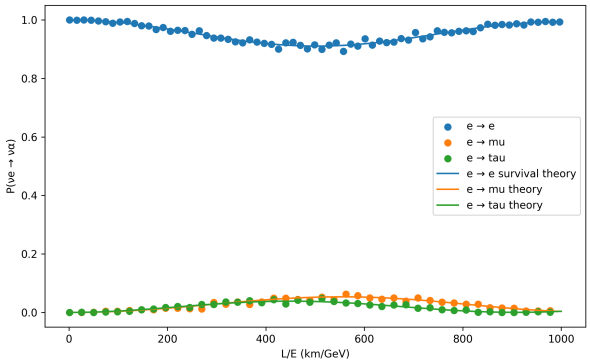


Figure 9.3: CP-incorporated $\nu_e \to \nu_\alpha$ appearance probabilities for small L/E, comparing the quantum-circuit simulation with the analytic three-flavor result. In this regime the atmospheric splitting dominates and the CP-odd term is small.

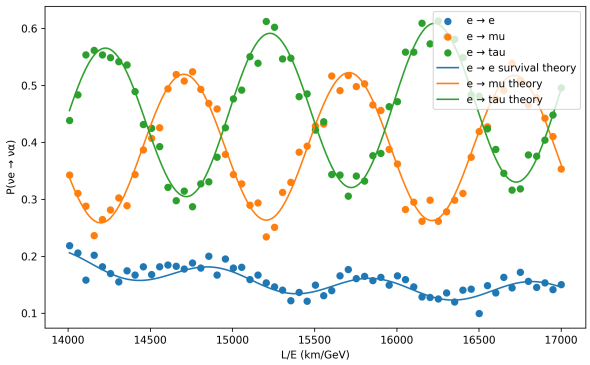


Figure 9.4: CP-incorporated $\nu_e \to \nu_\alpha$ appearance probabilities for large L/E. At large L/E both mass splittings contribute and the interference term proportional to $\sin\delta_{CP}$ visibly distorts the oscillation pattern relative to the CP-conserving case.

The quantum-circuit results track the analytic curves within the statistical (shot-noise) fluctuations expected from a finite number of measurements, confirming that the phase-gate-augmented decomposition faithfully realizes the complex PMNS matrix. Two features of the results are worth emphasizing. First, the CP-odd interference term of Eq. (9.2) is largest where the two oscillation frequencies associated with Δm^2_{21} and Δm^2_{31} interfere, which is why its effect is most pronounced in the large L/E regime of Fig. 9.4. Second, since the CP-odd term cancels in survival channels, the $\nu_e \to \nu_e$ probability is insensitive to δ_{CP} at this order; the sensitivity resides in the appearance channels $\nu_e \to \nu_\mu$ and $\nu_e \to \nu_\tau$, as observed in the simulation.

Finally, rerunning the identical circuit with all CP-dependent phase parameters negated ($\delta_{CP} \to -\delta_{CP}$) simulates the antineutrino channels, and the difference between the two runs reproduces the CP asymmetry of Eq. (9.6). This demonstrates that a two-qubit quantum processor can directly exhibit leptonic CP violation as a measurable difference between two circuit configurations, in complete analogy with the neutrino–antineutrino comparison performed by long-baseline experiments.

Chapter 10

Results and Discussion

The main objective of this thesis was not to perform large-scale hardware implementation, but to build the quantum-circuit formulation of neutrino oscillation in a fully explicit way and verify it through analytical derivations and Qiskit simulations. The key results of this work are summarized below:

- Starting from the ultra-relativistic limit of neutrino oscillation, the two-flavor formalism was derived explicitly and then extended in detail to the three-flavor case.
- The unitary operators relevant for the circuit construction were introduced systematically. In particular, the most general one-qubit unitary operator was derived, and it was shown explicitly how two phases may be removed by field rephasing. The phase gate S was then identified as a special case of the general U_3 gate.
- For the two-flavor case, the PMNS mixing matrix and the unitary time-evolution operator were rewritten explicitly in quantum-gate language. Qiskit circuits were then constructed for both survival and appearance channels. It was shown that measurement of the 0 state yields the electron-neutrino survival probability, while measurement of the 1 state yields the corresponding appearance probability, with the roles reversed for the muon-neutrino case.
- Theoretical probabilities and quantum-simulation results were plotted for the two-flavor case and shown to be in good agreement, establishing the consistency of the circuit construction with the standard oscillation formulae.
- For the three-flavor case, the PMNS matrix was constructed in detail in terms of quantum gates, and its Hermitian conjugate was derived explicitly. The unitary evolution operator was also rewritten in terms of S gates, and it was shown in detail how the choice of parameters allows the propagation part of the circuit to remain non-entangling, in agreement with the structure proposed in the literature.

- Qiskit circuits were then implemented for the three-flavor system, and the oscillation probabilities were studied for both short- and long-baseline regimes. It was shown explicitly that measurement outcomes 00, 01, and 10 correspond respectively to the electron-, muon-, and tau-neutrino probabilities.

- For the sterile-neutrino extension, the calculation was carried out explicitly to show how the enlarged unitary evolution operator necessarily generates entanglement. The corresponding circuit was reconstructed in detail in the quantum-computing framework.

- For the decoherence case, the circuit proposed in the literature was rederived and the relevant calculations were worked out explicitly, showing how decoherence can be incorporated within the circuit description of neutrino oscillation.

- For neutrino propagation in matter, the effective mixing angle and effective mass-squared difference were derived explicitly, yielding the modified oscillation parameters that enter the circuit construction.

Overall, this thesis provides a detailed derivation-based account of how neutrino oscillation, beginning from the standard analytical formalism, can be translated into the language of quantum circuits. Rather than only quoting known results, the emphasis here has been on explicitly deriving the required operators, constructing the circuits step by step, and demonstrating through Qiskit simulations that the quantum-circuit description reproduces the expected oscillation probabilities in both the two-flavor and three-flavor cases, while also extending naturally to sterile-neutrino and decoherence scenarios.

Chapter 11

Conclusion and Future Directions

In this thesis, we studied the quantum-circuit formulation of neutrino oscillation by following the framework proposed in the work of Arguelles *et al.* and deriving its main ingredients in detail. Starting from the standard neutrino-oscillation formalism in the ultra-relativistic limit, the two-flavor case was worked out explicitly and then extended to the three-flavor framework. The relevant unitary operators were introduced systematically, including the derivation of the most general one-qubit unitary operator and the explicit removal of unphysical phases by field rephasing. Within this formulation, the phase gate S was identified as a special case of the general U_3 gate.

For the two-flavor case, the PMNS mixing matrix and the unitary evolution operator were rewritten in quantum-gate language, and Qiskit circuits were implemented for both survival and appearance probabilities. It was shown explicitly how measurement of the computational basis states reproduces the corresponding oscillation probabilities. The theoretical results were then compared with the quantum simulations, and good agreement was obtained.

For the three-flavor case, the PMNS matrix was constructed in detail in terms of quantum gates, and its Hermitian conjugate was derived explicitly. The propagation operator was rewritten in terms of S gates, and it was shown in detail how the propagation part can be chosen so as not to introduce entanglement, in agreement with the structure presented in the literature. Qiskit circuits were then implemented for the three-flavor system, and the basis states 00, 01, and 10 were identified respectively with the electron-, muon-, and tau-neutrino probabilities. The resulting simulations were studied for both short- and long-baseline regimes.

The formalism was also extended to decoherence and matter effects. In the decoherence case, the circuit structure was reconstructed and the corresponding calculations were presented explicitly. For matter effects, the effective oscillation parameters in matter were derived and then incorporated into the same circuit framework. In this way, the thesis

has focused on giving a detailed derivation-based account of how the neutrino-oscillation formalism can be translated into the language of quantum circuits and verified through simulation.

Future work may proceed in several directions:

- The most immediate next step is to incorporate a non-zero CP-violating phase into the PMNS matrix and develop the corresponding quantum circuit. Unlike the cases considered in the present work, this requires keeping the genuinely complex phase structure of the mixing matrix and constructing a gate decomposition that reproduces it faithfully.
- A detailed study of the three-flavor circuit in the presence of CP violation would make it possible to examine explicitly how the additional phase modifies the oscillation probabilities and the corresponding measurement outcomes in the quantum-circuit framework.
- The decoherence framework can be extended further to include more general models, such as energy-dependent decoherence parameters or more realistic system-environment couplings.
- Matter effects may also be generalized beyond the constant-density approximation to spatially varying density profiles, which would require a piecewise implementation of the effective Hamiltonian.

Overall, this thesis has provided a detailed analytical and computational study of how known neutrino-oscillation formalisms can be expressed in the language of quantum circuits, while the incorporation of a non-zero CP-violating phase is now being pursued as the next original extension of this work.

Appendix A

Detailed Gate Decompositions

A.1 Code Availability

The computational work associated with this thesis is provided in an accompanying Jupyter notebook. This notebook contains the Qiskit code for the two-flavor circuits, the three-flavor probability plots, and the other circuit implementations and figures presented in this thesis. It therefore serves as a supplementary computational reference for the results discussed in the main text.

The notebook is cited in the bibliography and may also be accessed online through the corresponding repository or link[14].

A.2 Three-Flavor PMNS Decomposition

The six angles in the PMNS decomposition are obtained by solving a system of equations that match the matrix elements. For the standard parameterization, a numerical optimization yields the angles for a given set of mixing angles and CP phase. The explicit values are not essential for the circuit implementation; they can be precomputed and hard-coded into the gates. For completeness, we provide the mapping: given the 4×4 block-diagonal matrix $U = \mathrm{diag}(U^{3\times3}_{\mathrm{PMNS}}, 1)$, the decomposition into the sequence of two CNOTs and six U_3 gates follows from the Cartan decomposition of $SU(4)$. The specific angles can be found using the method described in [2]. The corresponding Qiskit implementations of this numerical fitting is available in the accompanying jupyter notebook[15].

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